Causal Survival Case Base SS v2

2026-07-07

Imports

```r
library("survival")
library("casebase")
library("sandwich")
library("ggplot2")
library("patchwork")
library("gt")
library("foreach")
library("doParallel")
library("doRNG")
```

Functions

Data Generation

```
generate.data = function(
    n,
    r,
    l,
    treatment.predictor.coefs,
    intercept,
    outcome.predictor.coefs,
    beta.A,
    censoring.limit,
    constant.HR = TRUE,
    shape = 1
  ) {

  # ---- Covariates (x1-x11) ----
  X1 <- rnorm(n)
  X2 <- rnorm(n)
  X3 <- runif(n)
  X4 <- runif(n)
  X5 <- rlnorm(n)
  X6 <- rlnorm(n)
  X7 <- rexp(n, rate = r)
  X8 <- rbinom(n, 1, 0.6)
  X9 <- rbinom(n, 1, 0.4)
  X10 <- rpois(n, lambda = l)

  X <- cbind(
    X1, X2, X3, X4, X5,
    X6, X7, X8, X9, X10
  )

  # ---- DbC ----
  stopifnot(length(treatment.predictor.coefs) == ncol(X))
  stopifnot(length(outcome.predictor.coefs) == ncol(X))
  stopifnot(all(is.finite(treatment.predictor.coefs)))
  stopifnot(all(is.finite(outcome.predictor.coefs)))

  # ---- Treatment (confounded by all covariates) ----
  weighted.covariates <- as.vector(X %*% treatment.predictor.coefs)
  linear.predictor.A <- (intercept) + weighted.covariates

  A <- rbinom(n, 1, plogis(linear.predictor.A))

  # ---- Outcome model (true causal effect of A) ----
  weighted.covariates <- as.vector(X %*% outcome.predictor.coefs)
  C <- runif(n, 0, censoring.limit) # random censoring

  linear.predictor.Y <- -1 + weighted.covariates
```

```r
  T.star <- numeric(n)

  if (constant.HR) {

    linear.predictor.Y <- linear.predictor.Y + (beta.A * A)

    T.star <- rweibull(
      n,
      shape = shape,
      scale = exp(-linear.predictor.Y / shape)
    )

  #time-dependent HR
  } else {
    # insert code for time dependent HR
    linear.predictor.Y <- NULL

    T.star <- NULL
  }

  time <- pmin(T.star, C) # observed event time
  Delta <- as.integer(T.star <= C) # event indicator

  # ---- Data Set Design ----
  data_set <- data.frame(
    time = as.numeric(time),
    Delta = Delta,
    A = A,
    X
  )

  return(data_set)
}
```

FSH: Modified Definition

```r
FSH <- function(
    formula,
    data,
    time,
    w = rep(1, nrow(data)),
    ratio = 100, ...
  ) {

  cl <- match.call()

  formula <- expand.dot.formula(
    formula,
    data = data
  )

  # Infer name of event variable from LHS of formula
  eventVar <- all.vars(formula[[2]])

  if (missing(time)) {
    varNames <- checkArgsTimeEvent(
      data = data,
      event = eventVar
    )

    timeVar <- varNames$time

  } else {
    timeVar <- time
  }

  typeEvents <- sort(unique(data[[eventVar]]))

  # Call sampleCaseBase if class is not cbData
  if (!inherits(data, "cbData")) {

    originalData <- as.data.frame(data)
    originalData$w <- w

    sampleData <- sampleCaseBase(
      originalData,
      timeVar,
      eventVar,
      comprisk = (length(typeEvents) > 2),
      ratio
    )

  } else {
    # If class is cbData we no longer have the original data
    originalData <- data.frame() # Empty data.frame
    sampleData <- data
  }
```

```r
  formula <- update(formula, ~ . + offset(offset))

  # Fit a binomial model because there are no competing risks
  fittingFunction <- function(formula) glm(
    formula,
    data = sampleData,
    family = binomial,
    weights = w
  )

  out <- fittingFunction(formula)

  # Save lower call for plot method
  out$lower_call <- out$call
  out$call <- cl

  out$originalData <- originalData
  out$typeEvents <- typeEvents
  out$timeVar <- timeVar
  out$eventVar <- eventVar

  # Count person-moment types
  num_pm <- table(sampleData[[eventVar]])
  out$num_cm <- num_pm[2]
  out$num_bm <- num_pm[1]
  out$formula <- formula

  # Reset offset for absolute risk estimation, but keep track of it
  out$offset <- out$data$offset
  out$data$offset <- 0

  # Add new class
  class(out) <- c("singleEventCB", class(out))

  return(out)
}
```

Expand Dot Formula

```
expand.dot.formula <- function(formula, data = NULL) {
  if (isTRUE("." %in% all.vars(formula))) {
    att <- attributes(formula)
    try.terms <- try(
      stats::terms(formula, data = data),
      silent = TRUE
    )
    if (!is(try.terms, "try-error")) {
      formula <- formula(try.terms)
    }
    attributes(formula) <- att
  }
  formula
}
```

RMST

```r
rmst.fn <- function(time, surv) {
  dt <- diff(time)
  mids <- (head(surv, -1) + tail(surv, -1)) / 2
  c(0, cumsum(mids * dt))
}
```

Refactor Data Structure

```r
refactor <- function(result) {
  store.by.var <- lapply(names(result[[1]]), function(x)
    sapply(result, `[[`, x)
  )
  names(store.by.var) <- names(result[[1]])
  return(store.by.var)
}
```

Get Marginal Estimates

```
get.marginal.estimates <- function(
    r, l,
    intercept,
    outcome.predictor.coefs,
    beta.A,
    censoring.limit,
    constant.HR
  ) {

  # generate data
  dat <- generate.data(
    n                          = 10000000,
    r                          = r,
    l                          = l,
    treatment.predictor.coefs = rep(0, 10),
    intercept                  = intercept,
    outcome.predictor.coefs   = outcome.predictor.coefs,
    beta.A                     = beta.A,
    censoring.limit            = censoring.limit,
    constant.HR                = constant.HR
  )

  # fit ms-cox model and extract psi
  ms.cox <- NULL
  psi <- NULL

  if (constant.HR) {
    ms.cox <- coxph(
      Surv(time, Delta) ~ A,
      data = dat
    )
    psi <- coef(ms.cox)["A"]

  } else {
    ms.cox <- coxph(
      Surv(time, Delta) ~ A * time,
      data = dat
    )
    psi <- NULL
  }

  ## extract counter-factual ms.cox survival curves
  sf.cox <- survfit(
    ms.cox,
    newdata = data.frame(A = c(0, 1))
  )

  s0.cox <- approx(
    sf.cox$time,
    sf.cox$surv[, 1],
    xout   = t.grid,
```

```
    method = "constant",
    rule   = 2
  )$y

  s1.cox <- approx(
    sf.cox$time,
    sf.cox$surv[, 2],
    xout   = t.grid,
    method = "constant",
    rule   = 2
  )$y

  # create output object
  output <- list(
    psi  = psi,
    rmst = rmst.fn(t.grid, s1.cox) - rmst.fn(t.grid, s0.cox)
  )

  return(output)
}
```

Intermediate Step

```r
intermediate.step <- function(
    r, l,
    intercept,
    outcome.predictor.coefs,
    beta.A,
    censoring.limit,
    constant.HR
  ) {

  cl <- parallel::makeCluster(2)
  doParallel::registerDoParallel(cl)
  on.exit(parallel::stopCluster(cl), add = TRUE)

  result <- foreach(
    i = 1:100,
    .packages = "survival",
    .export = c(
      "generate.data",
      "get.marginal.estimates",
      "t.grid", "rmst.fn"
    ),
    .options.RNG = seed
  ) %dorng% {
    get.marginal.estimates(
      r, l,
      intercept,
      outcome.predictor.coefs,
      beta.A,
      censoring.limit,
      constant.HR
    )
  }

  #refactor the results to efficiently store them then return
  return(refactor(result))
}
```

MS-Cox RMST SE

```
cox.rmst.list <- function(
    ms.cox, t.grid, cox.psi.se, survival.curves.data
  ) {

  # calculate counter-factual survival curves
  sf <- survfit(
    ms.cox,
    newdata = survival.curves.data
  )

  S0 <- approx(
    sf$time,
    sf$surv[, 1],
    t.grid,
    method = "constant",
    rule = 2
  )$y

  S1 <- approx(
    sf$time,
    sf$surv[, 2],
    t.grid,
    method = "constant",
    rule = 2
  )$y

  # point estimate
  cox.rmst <- rmst.fn(t.grid, S1) - rmst.fn(t.grid, S0)

  # delta-method gradient: g = int_0^tau S1 * log(S1) dt
  integrand <- ifelse(S1 > 0, S1 * log(S1), 0)
  g.cox <- rmst.fn(t.grid, integrand)
  cox.rmst.se <- abs(g.cox) * cox.psi.se

  return(list(
    cox.rmst = cox.rmst,
    cox.rmst.se = cox.rmst.se
  ))
}
```

Case Base RMST SE

```r
cb.rmst.list <- function(cb.ipw, cb.var, t.grid, constant.HR) {
  beta.hat  <- coef(cb.ipw)
  term.form <- delete.response(terms(cb.ipw))

  # survival S^{(a)}(t) built from the model's own design matrix
  cb.surv <- function(beta, a, grid) {
    nd <- data.frame(
      time = grid,
      A = a,
      offset = 0
    )

    X   <- model.matrix(term.form, data = nd)     # matches beta by construction
    X   <- X[, names(beta), drop = FALSE]         # align columns to coef order

    lambda <- exp(as.vector(X %*% beta))
    Lambda <- c(
      0,
      cumsum(
        diff(grid) * (head(lambda, -1) + tail(lambda, -1)) / 2
      )
    )
    exp(-Lambda)
  }

  rmst.diff <- function(beta) {
    rmst.fn(t.grid, cb.surv(beta, 1, t.grid)) -
    rmst.fn(t.grid, cb.surv(beta, 0, t.grid))
  }

  # calculate rmst
  cb.rmst <- rmst.diff(beta.hat)

  # numerical gradient of the RMST contrast wrt beta
  p <- length(beta.hat)
  m <- length(t.grid)

  grad <- matrix(NA_real_, nrow = m, ncol = p)
  colnames(grad) <- names(beta.hat)
  eps <- 1e-6

  for (j in seq_len(p)) {
    bp <- bm <- beta.hat
    bp[j] <- bp[j] + eps
    bm[j] <- bm[j] - eps

    grad[, j] <- (rmst.diff(bp) - rmst.diff(bm)) / (2 * eps)
  }

  # align covariance to the gradient names, then apply the delta method
  cb.var <- cb.var[colnames(grad), colnames(grad), drop = FALSE]
```

```r
  cb.rmst.se <- vapply(
    seq_len(m),
    function(i) {
      g <- grad[i, ]
      sqrt(drop(t(g) %*% cb.var %*% g))
    },
    numeric(1)
  )

  return(list(
    cb.rmst = cb.rmst,
    cb.rmst.se = cb.rmst.se
  ))
}
```

Run One Simulation

```
run.one.sim <- function(
    n, r, l,
    treatment.predictor.coefs,
    intercept,
    outcome.predictor.coefs,
    beta.A,
    censoring.limit,
    constant.HR,
    i
  ) {

  # generate data
  data.set <- generate.data(
      n                         = n,
      r                         = r,
      l                         = l,
      treatment.predictor.coefs = treatment.predictor.coefs,
      intercept                 = intercept,
      outcome.predictor.coefs   = outcome.predictor.coefs,
      beta.A                    = beta.A,
      censoring.limit           = censoring.limit,
      constant.HR               = constant.HR
  )

  # calculate propensity scores
  p.score <- fitted(
    glm(
      A ~ X1 + X2 + X3 + X4 + X5 +
        X6 + X7 + X8 + X9 + X10,
      data = data.set,
      family = binomial(link = "logit")
    )
  )

  A <- data.set$A

  p.A1 <- mean(A)

  # calculate stabilized ipws
  ipw.stabilized <- ifelse(
    A == 1,
    p.A1 / p.score,
    (1 - p.A1) / (1 - p.score)
  )

  # cox model
  ms.cox <- NULL
  cox.psi <- NULL
  cox.psi.se <- NULL

  if (constant.HR) {
```

```r
  ms.cox <- coxph(
    Surv(time, Delta) ~ A,
    data = data.set,
    weights = ipw.stabilized,
    robust = TRUE
  )
  ## extract hr and hr.se
  cox.psi <- coef(ms.cox)["A"]
  cox.psi.se <- sqrt(diag(vcov(ms.cox))["A"])

} else { # variable HR
  ms.cox <- coxph(
    Surv(time, Delta) ~ A * time,
    data = data.set,
    weights = ipw.stabilized,
    robust = TRUE
  )
  ## extract hr and hr.se
  cox.psi <- NULL
  cox.psi.se <- NULL
}

## extract counterfactual ms.cox survival curves
survival.curves.data <- data.frame(A = c(0, 1))

## calculate rmst and rmst.se
cox.rmst.list <- cox.rmst.list(
  ms.cox = ms.cox,
  t.grid = t.grid,
  cox.psi.se = cox.psi.se,
  survival.curves.data = survival.curves.data
)

cox.rmst <- cox.rmst.list$cox.rmst
cox.rmst.se <- cox.rmst.list$cox.rmst.se

# case-base model
fsh.ratio <- 100

## fit cb model
cb.ipw <- NULL
cb.psi <- NULL
cb.psi.se <- NULL
cb.var <- NULL

if (constant.HR) {
  cb.ipw <- FSH(
    Delta ~ A,
    data = data.set,
    ratio = fsh.ratio,
    w = ipw.stabilized
  )
  ## extract psi and psi.se
```

```r
    cb.psi <- coef(cb.ipw)["A"]
    cb.var <- vcovHC(cb.ipw, type = "HC0")
    cb.psi.se <- sqrt(diag(cb.var)["A"])

  } else {
    cb.ipw <- FSH(
      Delta ~ A * pspline(time, df = 4),
      data = data.set,
      ratio = fsh.ratio,
      w = ipw.stabilized
    )
    ## extract psi and psi.se
    cb.psi <- NULL
    cb.psi.se <- NULL
    cb.var <- NULL
  }

  ## calculate rmst and rmst.se
  cb.rmst.list <- cb.rmst.list(
    cb.ipw, cb.var, t.grid, constant.HR
  )

  cb.rmst <- cb.rmst.list$cb.rmst
  cb.rmst.se <- cb.rmst.list$cb.rmst.se


  output <- list(
    simulation.number = i,
    cox.psi = cox.psi,
    cox.psi.se = cox.psi.se,
    cb.psi = cb.psi,
    cb.psi.se = cb.psi.se,
    cox.rmst = cox.rmst,
    cox.rmst.se = cox.rmst.se,
    cb.rmst = cb.rmst,
    cb.rmst.se = cb.rmst.se
  )

  return(output)
}
```

Simulate

```r
simulate = function(
      n, r, l,
      treatment.predictor.coefs,
      intercept,
      outcome.predictor.coefs,
      beta.A,
      censoring.limit,
      constant.HR
  ) {

  # setup for parallel processing
  n.cores <- round(parallel::detectCores() * 0.6)
  cl <- parallel::makeCluster(n.cores)
  doParallel::registerDoParallel(cl)
  on.exit(parallel::stopCluster(cl), add = TRUE)

  # run the simulations
  result <- foreach(
    i = 1:1000,
    .packages = c("survival", "casebase"),
    .export = c(
      "run.one.sim",
      "generate.data",
      "vcovHC", "vcov",
      "FSH", "expand.dot.formula",
      "t.grid", "rmst.fn",
      "cox.rmst.list",
      "cb.rmst.list",
      "shape"
    ),
    .options.RNG = seed
  ) %dorng% {
    run.one.sim(
      n, r, l,
      treatment.predictor.coefs,
      intercept,
      outcome.predictor.coefs,
      beta.A,
      censoring.limit,
      constant.HR,
      i
    )
  }

  #refactor the results to efficiently store them
  return(refactor(result))
}
```

Plot Multiple RMST

```
plot.multiple.rmst <- function(titles, rmst.list, cols, main.title = "") {

  num.types <- length(rmst.list)

  if (main.title == "") {
    main.title <- paste(titles[1], "RMST")
    if (num.types > 1) {
      main.title <- "RMST Comparison"
    }
  }

  if (
    num.types == length(titles) &&
    num.types == length(cols) &&
    num.types > 0 &&
    num.types <= 5
  ) {

    avg.last.point <- mean(rmst.list[[1]][-nrow(rmst.list[[1]])])
    legend.placement <- "topright"

    if (avg.last.point < -0.3) {
      ylim <- c(min(rmst.list[[1]][-nrow(rmst.list[[1]])]), 0)
      legend.placement <- "bottomleft"

    } else if (avg.last.point > 0.3) {
      ylim <- c(0, max(rmst.list[[1]][-nrow(rmst.list[[1]])]))

    } else {
      ylim <- c(
        min(rmst.list[[1]][-nrow(rmst.list[[1]])]),
        max(rmst.list[[1]][-nrow(rmst.list[[1]])])
      )
    }

    alpha.cols <- adjustcolor(cols, alpha.f = 0.05)

    # First plot
    matplot(
      t.grid,
      rmst.list[[1]],
      type = "l",
      lty = 1,
      col = alpha.cols[1],
      xlab = "Time",
      ylab = "RMST",
      main = main.title
    )

    # Add remaining curves
    if (length(rmst.list) > 1) {
```

```r
      for (i in 2:num.types) {
        matplot(
          t.grid,
          rmst.list[[i]],
          type = "l",
          lty = 1,
          col = alpha.cols[i],
          add = TRUE
        )
      }
    }

    abline(h = 0, col = "black", lwd = 3, lty = 2)

    legend(
      legend.placement,
      legend = titles,
      col = cols[1:num.types],
      lwd = 2,
      lty = 1
    )
  } else {
    cat("error")
    if (length(titles) != num.types) {
      cat("mismatch")
    }
  }
}
```

Plot All RMST

```r
plot.all.rmst <- function(true.rmst, cox.rmst, cb.rmst, save.plots = FALSE) {
  # plot only true rmst
  if (save.plots) {
    pdf("true-rmst.pdf", width = 8, height = 5)
  }
  plot.multiple.rmst(
    titles = c("Marginal"),
    rmst.list = list(true.rmst),
    cols = "purple"
  )
  if (save.plots) {
    dev.off()
  }

  # plot cox.rmst with true rmst
  if (save.plots) {
    pdf("cox-rmst.pdf", width = 8, height = 5)
  }
  plot.multiple.rmst(
    titles = c(
      "MS-Cox",
      "Marginal"
    ),
    rmst.list = list(
      cox.rmst,
      true.rmst
    ),
    cols = c(
      "red",
      "black"
    ),
    main.title = "MS-Cox RMST"
  )
  if (save.plots) {
    dev.off()
  }

  # plot cb.rmst with true rmst
  if (save.plots) {
    pdf("cb-rmst.pdf", width = 8, height = 5)
  }
  plot.multiple.rmst(
    titles = c(
      "Case Base",
      "Marginal"
    ),
    rmst.list = list(
      cb.rmst,
      true.rmst
    ),
    cols = c(
```

```r
      "blue",
      "black"
    ),
    main.title = "Case Base RMST"
  )
  if (save.plots) {
    dev.off()
  }

  # plot rmst comparison
  if (save.plots) {
    pdf("all-rmst.pdf", width = 8, height = 5)
  }
  plot.multiple.rmst(
    titles = c(
      "MS-Cox",
      "Case Base",
      "Marginal"
    ),
    rmst.list = list(
      cox.rmst,
      cb.rmst,
      true.rmst
    ),
    cols = c(
      "red",
      "blue",
      "black"
    )
  )
  if (save.plots) {
    dev.off()
  }
}
```

RMST Summary

```r
get.rmst.summary <- function(
    tau.points,
    rmst.matrix,
    rmst.se.matrix,
    marginal.rmst.means,
    alpha = 0.05
) {

  n <- ncol(rmst.matrix)
  z <- qnorm(1 - alpha / 2)

  ## Mean
  rmst.means <- rowMeans(
    rmst.matrix[tau.points, ]
  )

  ## Empirical SD
  rmst.sd <- apply(
    rmst.matrix[tau.points, ],
    1,
    sd
  )

  ## Mean estimated SE
  rmst.se.mean <- rowMeans(
    rmst.se.matrix[tau.points, ]
  )

  ## Bias
  rmst.biases <- rmst.means - marginal.rmst.means

  ## Bias MCSE
  rmst.bias.mcse <- rmst.sd / sqrt(n)

  ## RMSE
  rmst.rmse <- sqrt(
    rowMeans(
      (rmst.matrix[tau.points, ] - marginal.rmst.means)^2
    )
  )

  ## Coverage
  lower <- rmst.matrix[tau.points, ] -
    z * rmst.se.matrix[tau.points, ]

  upper <- rmst.matrix[tau.points, ] +
    z * rmst.se.matrix[tau.points, ]

  coverage <- rowMeans(
    lower <= marginal.rmst.means &
    upper >= marginal.rmst.means
```

```
  )

  ## Coverage MCSE
  coverage.mcse <- sqrt(
    coverage * (1 - coverage) / n
  )

  ## Store results
  k <- length(tau.points)

  rmst.summary <- matrix(NA_real_, nrow = k, ncol = 8)
  colnames(rmst.summary) <- c(
    "mean",
    "sd",
    "se.mean",
    "bias",
    "bias.mcse",
    "rmse",
    "coverage",
    "coverage.mcse"
  )

  rmst.summary[, "mean"] <- rmst.means
  rmst.summary[, "sd"] <- rmst.sd
  rmst.summary[, "se.mean"] <- rmst.se.mean
  rmst.summary[, "bias"] <- rmst.biases
  rmst.summary[, "bias.mcse"] <- rmst.bias.mcse
  rmst.summary[, "rmse"] <- rmst.rmse
  rmst.summary[, "coverage"] <- coverage
  rmst.summary[, "coverage.mcse"] <- coverage.mcse

  return(rmst.summary)
}
```

Output RMST Statistics Summary

```
print.rmst.summary <- function(rmst.summary, tau.values, model) {

  cat("\n========================================\n")
  cat(model, "\n")
  cat("========================================\n")

  cat("\nRestricted Mean Survival Time Summary\n")
  cat("----------------------------------------\n")

  for (i in seq_along(tau.values)) {

    cat("\nTau =", tau.values[i], "\n")
    cat("----------------------------------------\n")

    cat("Mean RMST:", rmst.summary[i, "mean"], "\n")
    cat("Bias:", rmst.summary[i, "bias"],
        " | MC-SE:", rmst.summary[i, "bias.mcse"], "\n")
    cat("RMSE:", rmst.summary[i, "rmse"], "\n")
    cat("Mean Estimated SE:", rmst.summary[i, "se.mean"],
        " | Empirical SD:", rmst.summary[i, "sd"], "\n")
    cat("Coverage:", rmst.summary[i, "coverage"],
        " | MC-SE:", rmst.summary[i, "coverage.mcse"], "\n")
  }

  cat("\n")
}
```

Output True RMST

```r
print.true.rmst <- function(true.rmst, tau.points, tau.values) {
  cat("\n========================================\n")
  cat("\nTrue Mean Restricted Mean Survival Time\n")
  cat("========================================\n")

  for (i in seq_along(tau.values)) {
    cat(tau.values[i], ":", mean(true.rmst[tau.points[i], ]), "\n")
  }
  cat("\n")
}
```

Plot HR

```
plot.hr <- function(true.hr, cox.hr, cb.hr, save.plot = FALSE) {

  if (save.plot) {
    pdf("hr.pdf", width = 8, height = 5)
  }

  hist(
    cox.hr,
    breaks = 50,
    col = rgb(1, 0, 0, 0.5),    # semi-transparent red
    border = "red",
    xlab = "Hazard Ratio",
    main = "Hazard Ratio Comparison"
  )

  hist(
    cb.hr,
    breaks = 50,
    col = rgb(0, 0, 1, 0.5),    # semi-transparent blue
    border = "blue",
    add = TRUE
  )

  abline(v = true.hr, col = "black", lwd = 3, lty = 2)

  legend(
    "topright",
    legend = c("MS-Cox HR", "Case Base HR", "True HR"),
    fill = c(rgb(1, 0, 0, 0.5), rgb(0, 0, 1, 0.5), NA),
    border = c("red", "blue", NA),
    lty = c(NA, NA, 2),
    lwd = c(NA, NA, 3),
    col = c(NA, NA, "black")
  )

  if (save.plot) {
    dev.off()
  }
}
```

Plot HR Errors

```r
plot.hr.errors <- function(true.hr, cox.hr, cb.hr, save.plot = FALSE) {
  cox.hr.error <- cox.hr - true.hr
  cb.hr.error <- cb.hr - true.hr

  if (save.plot) {
    pdf("hr-errors.pdf", width = 8, height = 5)
  }

  hist(
    cox.hr.error,
    breaks = 50,
    col = rgb(1, 0, 0, 0.5),   # semi-transparent red
    border = "red",
    xlab = "HR Errors",
    main = "HR Errors Comparison"
  )

  hist(
    cb.hr.error,
    breaks = 50,
    col = rgb(0, 0, 1, 0.5),   # semi-transparent blue
    border = "blue",
    add = TRUE
  )

  abline(v = 0, col = "black", lwd = 3, lty = 2)

  legend(
    "topleft",
    legend = c("MS-Cox HR Error", "Case Base HR Error"),
    fill = c(rgb(1, 0, 0, 0.5), rgb(0, 0, 1, 0.5)),
    border = c("red", "blue")
  )

  if (save.plot) {
    dev.off()
  }
}
```

HR Summary

```r
get.hr.summary <- function(
    true.hr,
    psi.vector,
    se.vector,
    alpha = 0.05,
    power = TRUE
  ) {
  n <- length(psi.vector)
  z <- qnorm(1 - alpha / 2)
  hr.vector <- exp(psi.vector)

  hr.mean <- exp(mean(psi.vector))
  hr.emp.sd <- sd(hr.vector)
  hr.bias <- hr.mean - true.hr
  hr.bias.mcse <- hr.emp.sd  / sqrt(n)
  hr.mse <- mean((hr.vector - true.hr)^2)
  hr.rmse <- sqrt(hr.mse)
  hr.se.mean <- mean(se.vector)
  hr.ci.lower <- exp(psi.vector - z * se.vector)
  hr.ci.upper <- exp(psi.vector + z * se.vector)
  t1.error.or.power <- mean(hr.ci.lower > 1 | hr.ci.upper < 1)
  hr.coverage <- mean(hr.ci.lower <= true.hr & true.hr <= hr.ci.upper)
  hr.coverage.mcse <- sqrt(hr.coverage * (1 - hr.coverage) / n)

  # store in a list
  results <- list(
    summary = c(
      mean                = hr.mean,
      emp.sd              = hr.emp.sd,
      bias                = hr.bias,
      bias.mcse           = hr.bias.mcse,
      mse                 = hr.mse,
      rmse                = hr.rmse,
      se.mean             = hr.se.mean,
      t1.error.or.power   = t1.error.or.power,
      coverage            = hr.coverage,
      coverage.mcse       = hr.coverage.mcse
    ),
    ci = cbind(
      lower = hr.ci.lower,
      upper = hr.ci.upper
    )
  )

  return(results)
}
```

Output HR Statistics Summary

```r
print.hr.summary <- function(hr.stats.summary, model) {

  cat("\n========================================\n")
  cat(model, "\n")
  cat("========================================\n")

  cat("\nHazard Ratio Summary\n")
  cat("----------------------------------------\n")

  cat("Mean HR:", hr.stats.summary["mean"], "\n")
  cat("Bias:", hr.stats.summary["bias"],
      " | MC-SE:", hr.stats.summary["bias.mcse"], "\n")
  cat("MSE:", hr.stats.summary["mse"], "\n")
  cat("RMSE:", hr.stats.summary["rmse"], "\n")
  cat("Mean Estimated SE:", hr.stats.summary["se.mean"],
      " | Empirical SD:", hr.stats.summary["emp.sd"], "\n")
  cat("Type I Error/Power:", hr.stats.summary["t1.error.or.power"], "\n")
  cat("Coverage:", hr.stats.summary["coverage"],
      " | MC-SE:", hr.stats.summary["coverage.mcse"], "\n")

  cat("\n")
}
```

Output True HR

```r
print.true.hr <- function(true.hr) {
  cat("\n========================================\n")
  cat("True HR:", true.hr, "\n")
  cat("========================================\n")
}
```

Summarize and Print HR Statistics

```
summarize.print.hr <- function(
    true.hr,
    cox.psi, cox.psi.se,
    cb.psi, cb.psi.se
  ) {

  cox.psi.stats <- get.hr.summary(
    true.hr,
    cox.psi,
    cox.psi.se
  )

  cb.psi.stats <- get.hr.summary(
    true.hr,
    cb.psi,
    cb.psi.se
  )

  print.hr.summary(
    cox.psi.stats$summary,
    "MS-Cox"
  )
  print.hr.summary(
    cb.psi.stats$summary,
    "Case Base"
  )

  return(list(
    cox.psi.stats,
    cb.psi.stats
  ))
}
```

Plot HR CI

```
plot.hr.ci <- function(
    true.value,
    title,
    ci.list,
    hr.list,
    colour = "red",
    coverage = 0,
    seed = 123
  ) {

  set.seed(seed)
  x = 1:100
  sampled.indices <- sample(1:length(hr.list), 100)

  plot <- ggplot() +
    geom_ribbon(
      aes(
        x = x,
        ymin = ci.list[[1]][sampled.indices],
        ymax = ci.list[[2]][sampled.indices]
      ),
      fill = colour,
      alpha = 0.5,
      show.legend = FALSE
    ) +
    geom_line(
      aes(
        x = x,
        y = hr.list[sampled.indices],
        colour = title
      )
    ) +
    geom_hline(
      aes(
        yintercept = true.value,
        colour = "True HR",
        linetype = "True HR"
      )
    ) +
    geom_hline(
      aes(
        yintercept = 1,
        colour = "Null value",
        linetype = "Null value"
      )
    ) +
    scale_colour_manual(
      name = "",
      values = c(
        setNames(colour, title),
        "True HR" = "black",
```

```
        "Null value" = "purple"
      )
    ) +
    scale_linetype_manual(
      name = "",
      values = c(
        title = "solid",
        "True HR" = "dashed",
        "Null value" = "dotted"
      )
    ) +
    labs(
      x = "Simulation",
      y = "Hazard ratio",
      title = paste(title, "HR Confidence Intervals - Coverage:", coverage)
    ) +
    theme_bw()

  return(plot)
}
```

Plot All HR CIs

```
plot.all.hr.ci <- function(
    true.hr,
    cox.hr, cox.hr.stats,
    cb.hr, cb.hr.stats,
    seed = 123,
    save.plot = FALSE
  ) {

  cox.hr.ci.plot <- plot.hr.ci(
    true.hr,
    title = "MS-Cox",
    list(
      cox.hr.stats$ci[, "lower"],
      cox.hr.stats$ci[, "upper"]
    ),
    cox.hr,
    colour = "red",
    coverage = cox.hr.stats$summary["coverage"],
    seed = seed
  )

  cb.hr.ci.plot <- plot.hr.ci(
    true.hr,
    title = "Case Base",
    list(
      cb.hr.stats$ci[, "lower"],
      cb.hr.stats$ci[, "upper"]
    ),
    cb.hr,
    colour = "blue",
    coverage = cb.hr.stats$summary["coverage"],
    seed = seed
  )

  combined_plot <- (cox.hr.ci.plot / cb.hr.ci.plot) +
  plot_layout(guides = "collect")

  if (save.plot) {
    ggsave(
      "hr-ci.pdf",
      combined_plot,
      width = 8,
      height = 8
    )
  }
}
```

Run a complete Simulation Study Function

```
run.complete.ss <- function(
    n,
    beta.A,
    censoring.limit,
    marginal.estimates,
    generate.plots = FALSE,
    save.plots = FALSE
  ) {

  need.plots <- generate.plots || save.plots

  ## Run the simulation study
  ss.out <- simulate(
    n, r, l,
    treatment.predictor.coefs,
    intercept,
    outcome.predictor.coefs,
    beta.A,
    censoring.limit,
    constant.HR
  )

  ## Restricted Mean Survival Time (RMST)

  if (need.plots) {
    ### All RMST Plots
    plot.all.rmst(
      marginal.estimates$rmst,
      ss.out$cox.rmst,
      ss.out$cb.rmst,
      save.plots = save.plots
    )
  }

  ### Summarize RMST Statistics
  tau.points <- which(
    (t.grid != 0) &
    (t.grid == floor(t.grid))
  )

  marginal.rmst.means <- rowMeans(
    marginal.estimates$rmst[tau.points, ]
  )
  marginal.rmst.sd <- apply(
    marginal.estimates$rmst[tau.points, ],
    1, sd
  )

  cox.rmst.summary <- get.rmst.summary(
    tau.points,
    ss.out$cox.rmst,
```

```
  ss.out$cox.rmst.se,
  marginal.rmst.means
)

cb.rmst.summary <- get.rmst.summary(
  tau.points,
  ss.out$cb.rmst,
  ss.out$cb.rmst.se,
  marginal.rmst.means
)

tau.values <- seq(1, 4, by = 1)

print.true.rmst(
  marginal.estimates$rmst,
  tau.points,
  tau.values
)

print.rmst.summary(
  cox.rmst.summary,
  tau.values,
  "MS-Cox RMST"
)

print.rmst.summary(
  cb.rmst.summary,
  tau.values,
  "Case Base RMST"
)

## Hazard Ratio (HR)

### Extract the True HR
true.hr <- exp(mean(marginal.estimates$psi))

if (need.plots) {
  ### Plot the distribution of the True HR
  if (save.plots) {
    pdf("true-hr.pdf", width = 8, height = 5)
  }
  hist(
    exp(marginal.estimates$psi),
    breaks = 30,
    xlab = "HR",
    main = "Marginal HR"
  )
  if (save.plots) {
    dev.off()
  }

  ### Visualize the Hazard Ratios
  plot.hr(
```

```r
      true.hr,
      exp(ss.out$cox.psi),
      exp(ss.out$cb.psi),
      save.plot = save.plots
    )

    ### Visualize the Hazard Ratio Errors
    plot.hr.errors(
      true.hr,
      exp(ss.out$cox.psi),
      exp(ss.out$cb.psi),
      save.plot = save.plots
    )
  }

  print.true.hr(true.hr)

  ### Summarize and Print HR Statistics
  all.hr.stats <- summarize.print.hr(
    true.hr,
    ss.out$cox.psi, ss.out$cox.psi.se,
    ss.out$cb.psi, ss.out$cb.psi.se
  )

  if (need.plots) {
    ### Plot All HR CIs
    plot.all.hr.ci(
      true.hr,
      exp(ss.out$cox.psi), all.hr.stats[[1]],
      exp(ss.out$cb.psi), all.hr.stats[[2]],
      seed = seed,
      save.plot = save.plots
    )
  }
}
```

Simulation Study

Data Generation Parameters

```r
seed <- 123

tau.max <- 4
t.grid <- seq(0, tau.max, by = 0.1)

r = 2
l = 1.5

treatment.predictor.coefs <- c(
  0.7,  -0.4,  0.5,  0.2, 0.6,
  -0.3,  0.1,  -0.2, 0.4, -0.5
)

outcome.predictor.coefs <- c(
  0.6,  -0.3,  0.4,  0.1, 0.5,
  -0.2,  0.05,  -0.1, 0.3, -0.4
)

intercept <- 0.3
constant.HR <- TRUE
shape = 1
```

Setting 1: beta.A = 0, n = 1000 (Null)

```
n1 = 1000
beta.A1 <- 0
censoring.limit.1 <- 5
```

Setting 2: beta.A = 1, n = 1000 (Strong)

```
n2 = 1000
beta.A2 <- 1
censoring.limit.2 <- 5
```

Setting 3: beta.A = 0.5, n = 1000 (Moderate)

```
n3 = 1000
beta.A3 <- 0.5
censoring.limit.3 <- 5
```

Setting 4: beta.A = 0.5, n = 500 (Moderate)

```
n4 = 500
beta.A4 <- 0.5
censoring.limit.4 <- 5
```

Setting 5: beta.A = 0.5, n = 250 (Moderate)

```
n5 = 250
beta.A5 <- 0.5
censoring.limit.5 <- 5
```

Setting 6: beta.A = 0.5, n = 1000 (Heavy Censoring)

```
n6 = 1000
beta.A6 <- 0.5
censoring.limit.6 <- 3
```

Extract the marginal estimates for all settings

Moderate Beta

```
true.estimates.moderate <- intermediate.step(
  r, l,
  intercept,
  outcome.predictor.coefs,
  beta.A3,
  censoring.limit.1,
  constant.HR
)
```

Null Beta

```
true.estimates.null <- intermediate.step(
  r, l,
  intercept,
  outcome.predictor.coefs,
  beta.A1,
  censoring.limit.1,
  constant.HR
)
```

Strong Beta

```
true.estimates.strong <- intermediate.step(
  r, l,
  intercept,
  outcome.predictor.coefs,
  beta.A2,
  censoring.limit.1,
  constant.HR
)
```

Time Dependent HR Settings

Monotone Inc/Dec Linear

Monotone Inc/Dec Quadratic

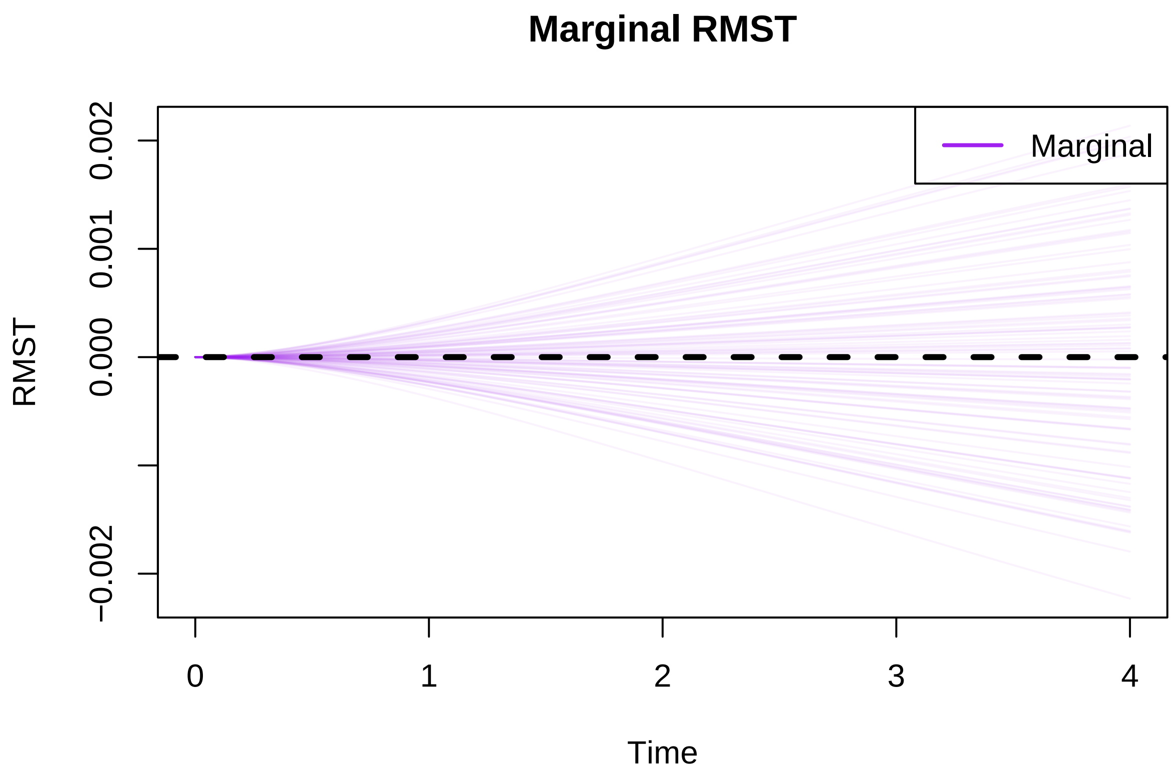
Quadratic Both Inc/Dec

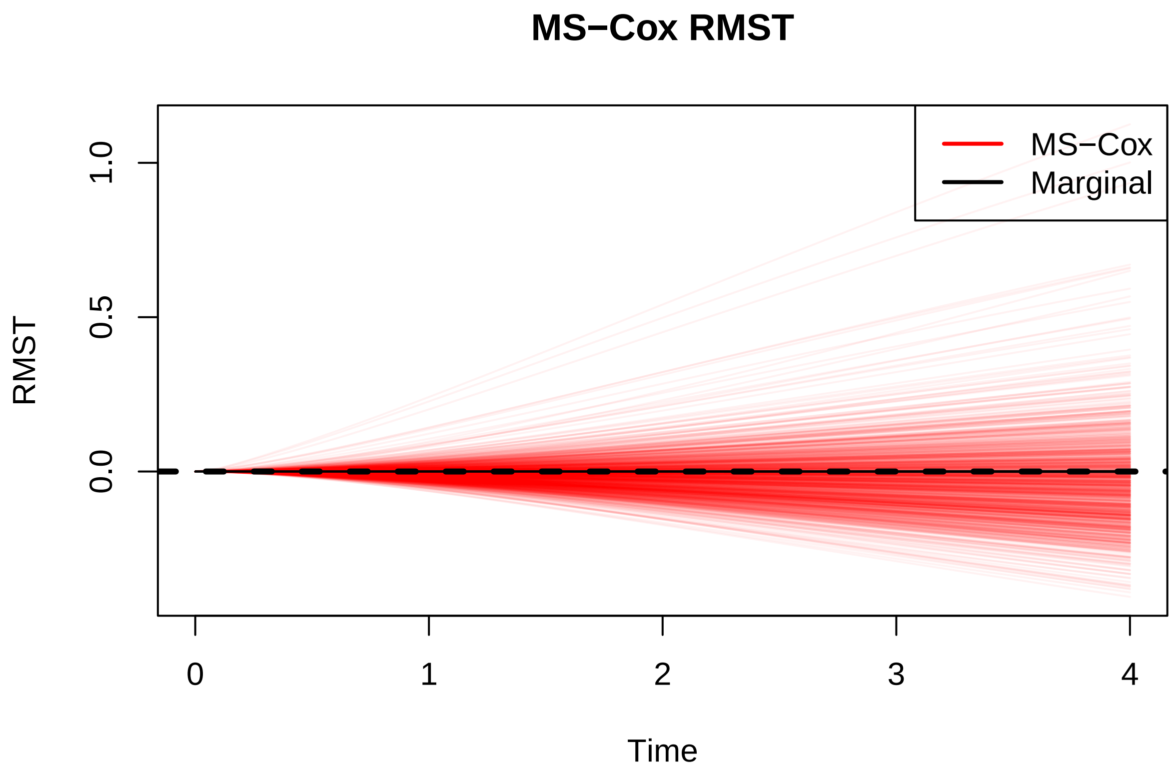
Weibull

Gompertz

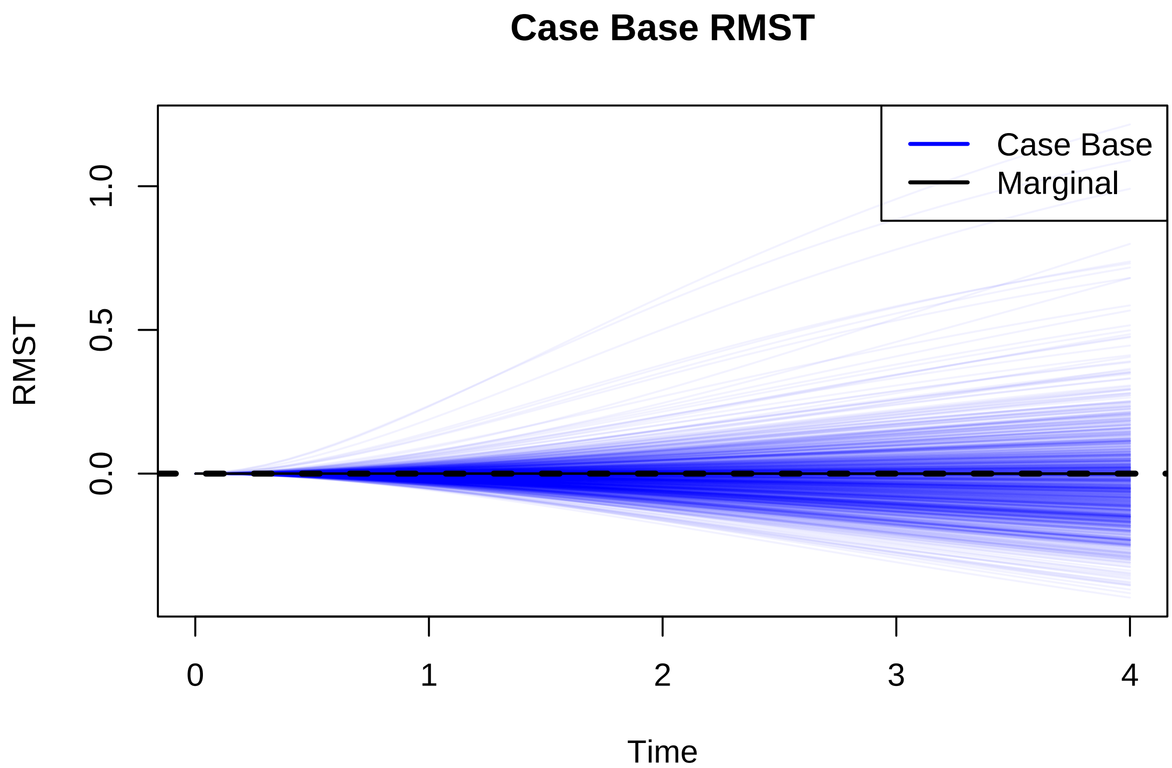
Run Simulation Study: Setting 1

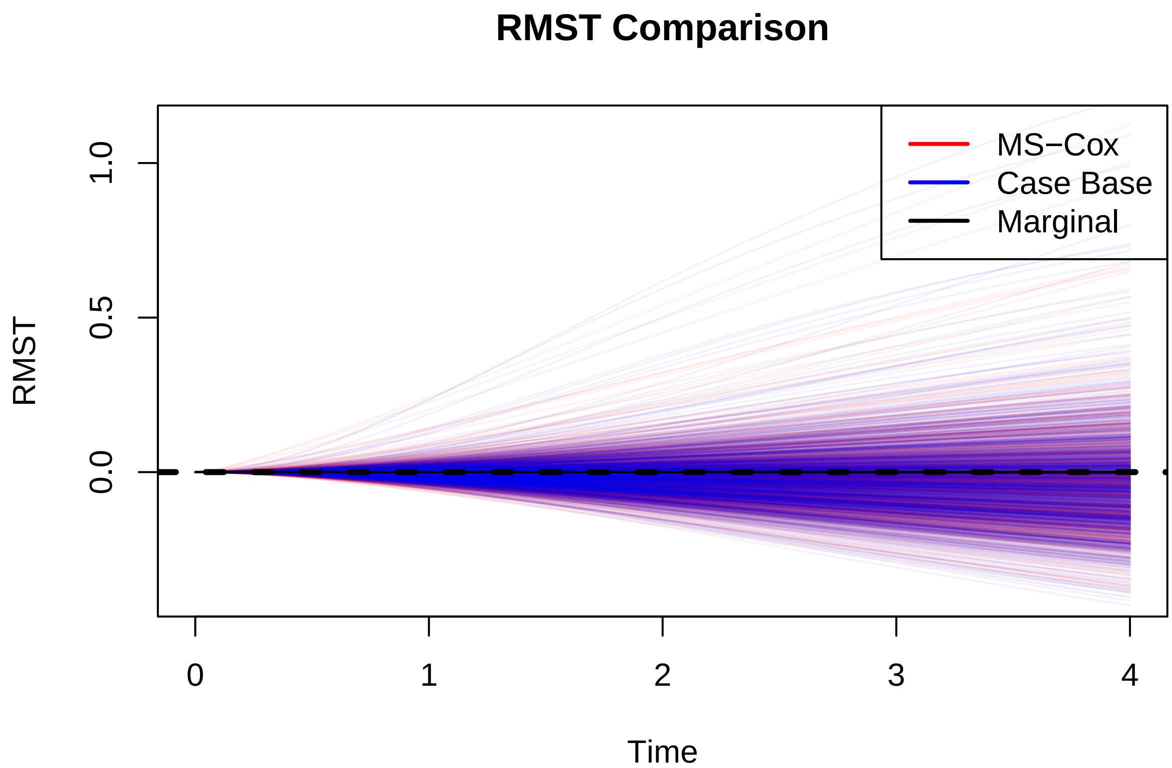
```
run.complete.ss(
  n1, beta.A1,
  censoring.limit.1,
  true.estimates.null,
  generate.plots = TRUE
)
```





MS−Cox RMST
MS−Cox
Marginal
RMST
1.0
0.5
0.0
0
1
2
3
4
Time

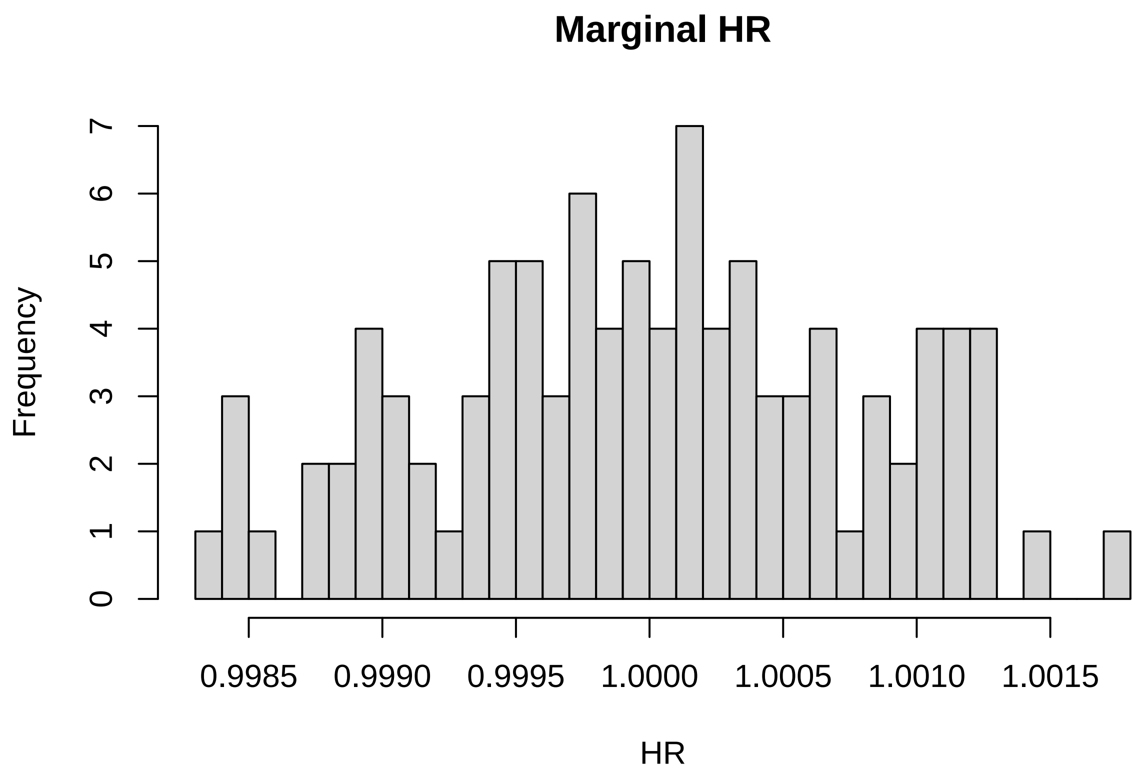

Case Base RMST
Case Base
Marginal
RMST
1.0
0.5
0.0
0
1
2
3
4
Time

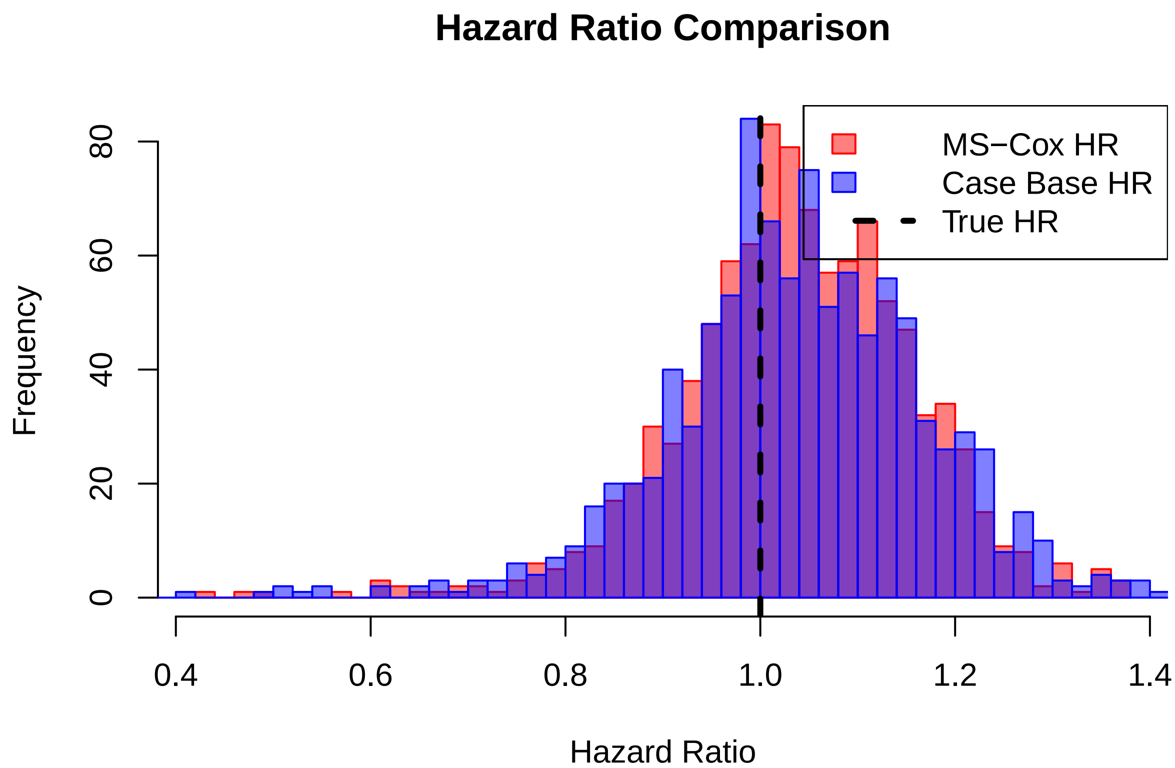


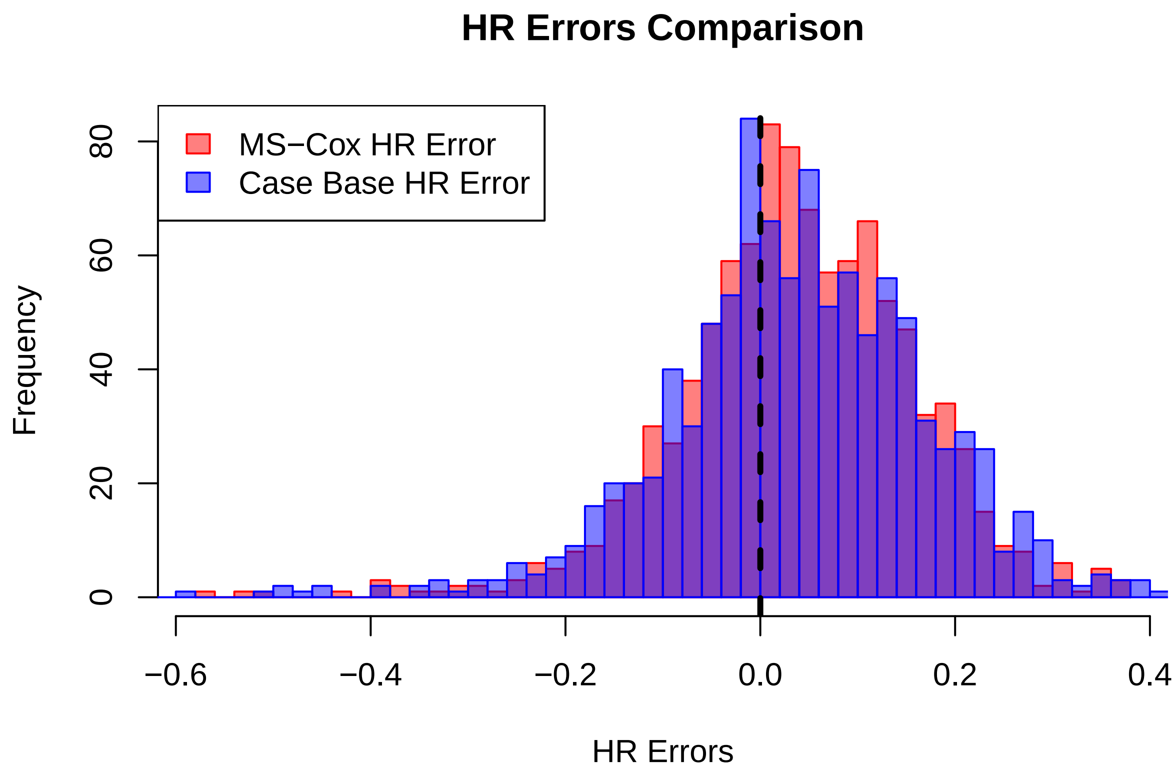

```
##
## ========================================
##
## True Mean Restricted Mean Survival Time
## ========================================
## 1 : 1.157282e-06
## 2 : 3.043589e-06
## 3 : 5.042438e-06
## 4 : 7.000952e-06
##
##
## ========================================
## MS-Cox RMST
## ========================================
##
## Restricted Mean Survival Time Summary
## ----------------------------------------
##
## Tau = 1
## ----------------------------------------
## Mean RMST: -0.004887401
## Bias: -0.004888558  | MC-SE: 0.0008914038
## RMSE: 0.02859553
## Mean Estimated SE: 0.02739353  | Empirical SD: 0.02818866
## Coverage: 0.97  | MC-SE: 0.005394442
##
```

```
## Tau = 2
## ----------------------------------------
## Mean RMST: -0.01450225
## Bias: -0.0145053  | MC-SE: 0.002255555
## RMSE: 0.07275195
## Mean Estimated SE: 0.07238266  | Empirical SD: 0.07132693
## Coverage: 0.968  | MC-SE: 0.005565609
##
## Tau = 3
## ----------------------------------------
## Mean RMST: -0.02511931
## Bias: -0.02512435  | MC-SE: 0.003694867
## RMSE: 0.1194555
## Mean Estimated SE: 0.1202928  | Empirical SD: 0.1168419
## Coverage: 0.967  | MC-SE: 0.005648982
##
## Tau = 4
## ----------------------------------------
## Mean RMST: -0.03556757
## Bias: -0.03557457  | MC-SE: 0.005100574
## RMSE: 0.1650921
## Mean Estimated SE: 0.1671634  | Empirical SD: 0.1612943
## Coverage: 0.966  | MC-SE: 0.005730969
##
##
## ========================================
## Case Base RMST
## ========================================
##
## Restricted Mean Survival Time Summary
## ----------------------------------------
##
## Tau = 1
## ----------------------------------------
## Mean RMST: -0.003672476
## Bias: -0.003673633  | MC-SE: 0.0008060566
## RMSE: 0.0257405
## Mean Estimated SE: 0.02237252  | Empirical SD: 0.02548975
## Coverage: 0.943  | MC-SE: 0.007331507
##
## Tau = 2
## ----------------------------------------
## Mean RMST: -0.01192917
## Bias: -0.01193222  | MC-SE: 0.002352874
## RMSE: 0.07531839
## Mean Estimated SE: 0.06740808  | Empirical SD: 0.07440442
## Coverage: 0.943  | MC-SE: 0.007331507
##
## Tau = 3
## ----------------------------------------
## Mean RMST: -0.02166483
## Bias: -0.02166987  | MC-SE: 0.003971813
## RMSE: 0.1273935
## Mean Estimated SE: 0.1161373  | Empirical SD: 0.1255998
```

```
## Coverage: 0.943  | MC-SE: 0.007331507
##
## Tau = 4
## ---------------------------------------
## Mean RMST: -0.03109219
## Bias: -0.03109919  | MC-SE: 0.005418617
## RMSE: 0.1740667
## Mean Estimated SE: 0.1605116  | Empirical SD: 0.1713517
## Coverage: 0.942  | MC-SE: 0.007391617
```




Hazard Ratio Comparison
MS−Cox HR
Case Base HR
True HR
Frequency
0
20
40
60
80
0.4
0.6
0.8
1.0
1.2
1.4
Hazard Ratio

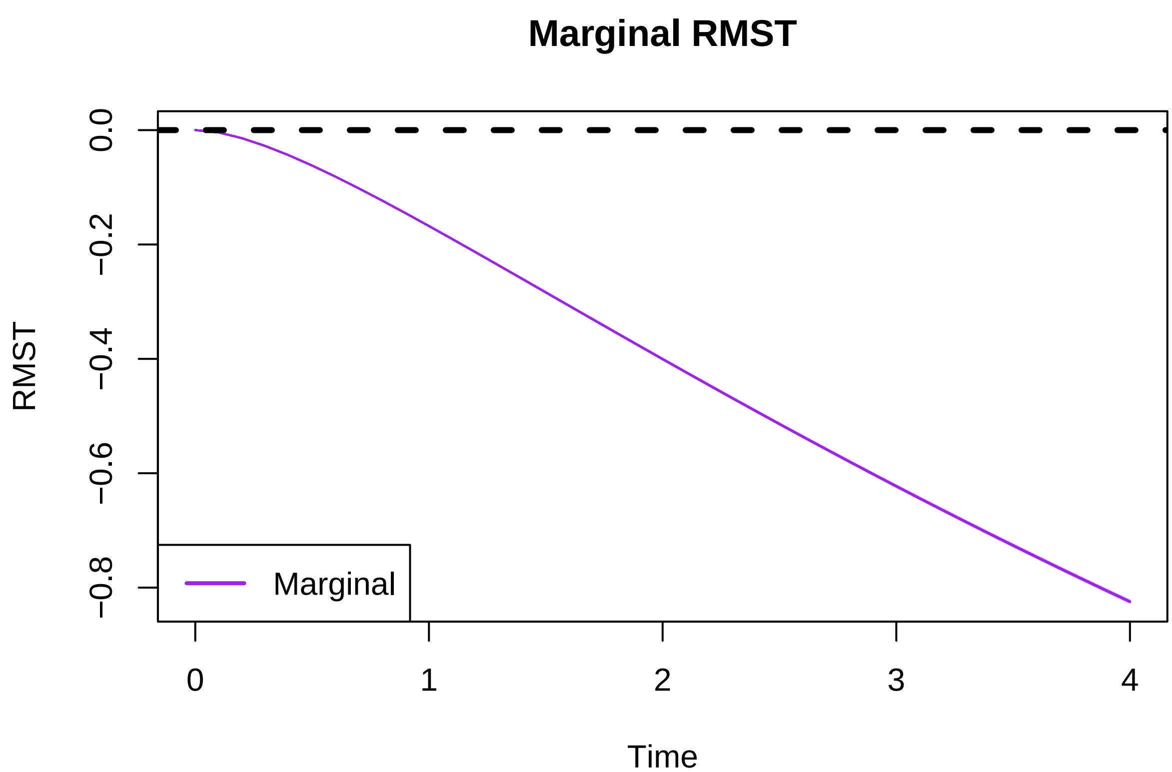


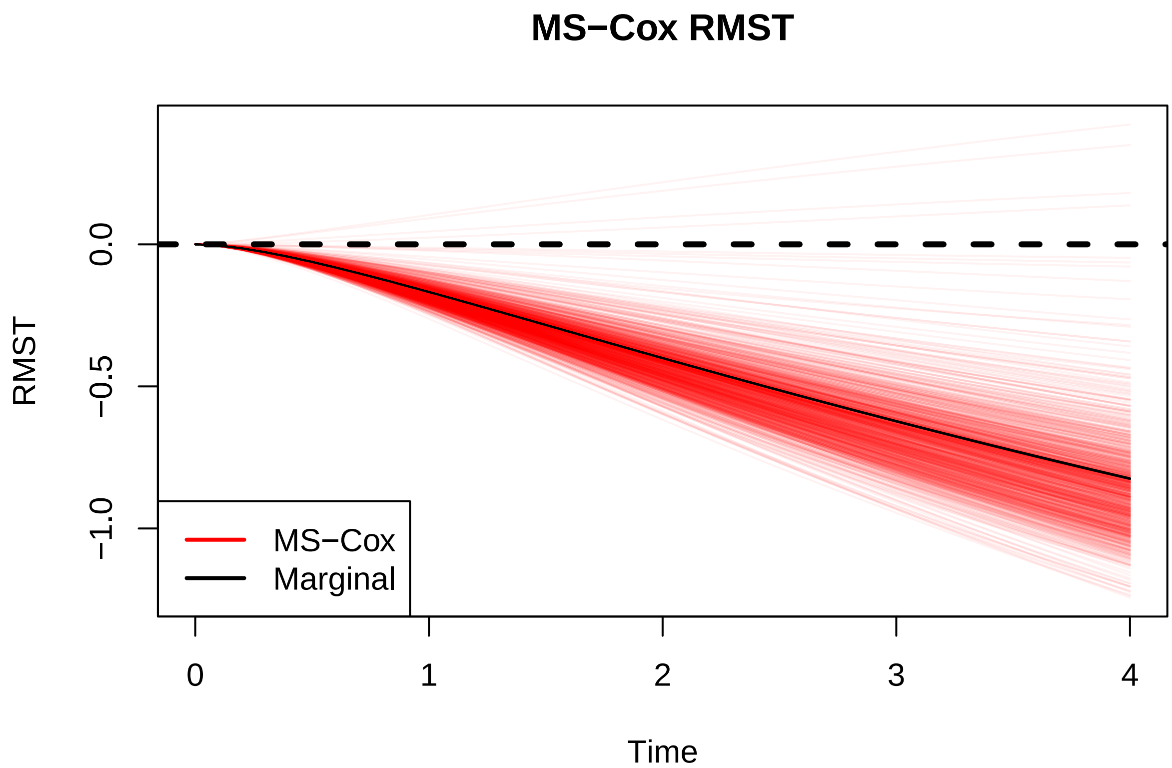

```
##
## ========================================
## True HR: 0.9999946
## ========================================
##
## ========================================
## MS-Cox
## ========================================
##
## Hazard Ratio Summary
## ----------------------------------------
## Mean HR: 1.029249
## Bias: 0.0292545  | MC-SE: 0.003848746
## MSE: 0.01616755
## RMSE: 0.1271517
## Mean Estimated SE: 0.1312366  | Empirical SD: 0.121708
## Type I Error/Power: 0.034
## Coverage: 0.966  | MC-SE: 0.005730969
##
##
## ========================================
## Case Base
## ========================================
##
## Hazard Ratio Summary
## ----------------------------------------
```

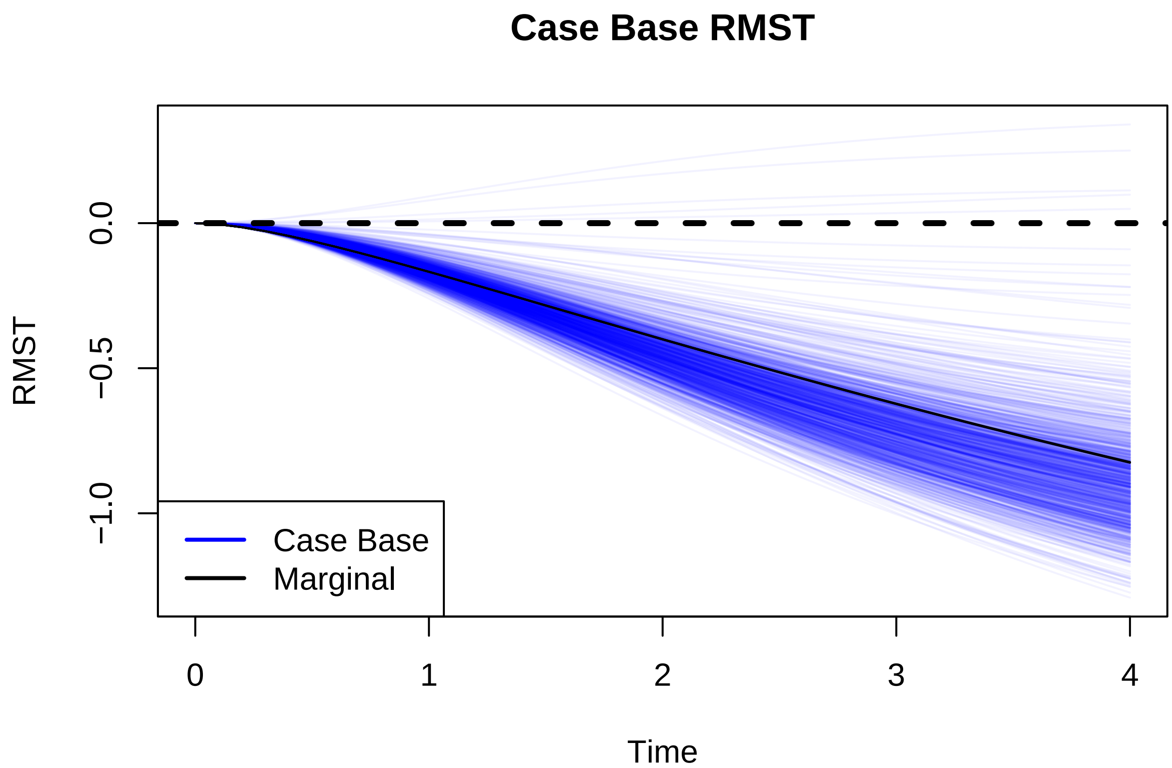
```
## Mean HR: 1.025946
## Bias: 0.02595193  | MC-SE: 0.004359047
## MSE: 0.0202971
## RMSE: 0.1424679
## Mean Estimated SE: 0.1362137  | Empirical SD: 0.1378452
## Type I Error/Power: 0.054
## Coverage: 0.946  | MC-SE: 0.007147307
```

Run Simulation Study: Setting 2

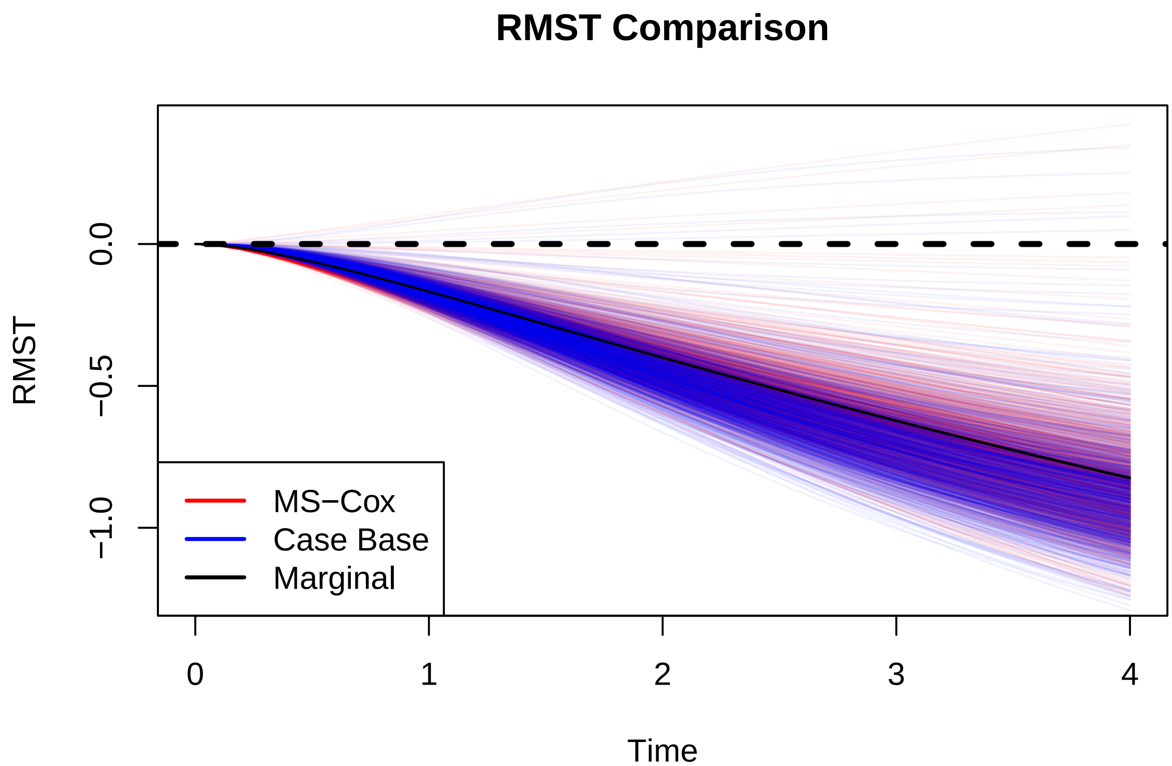
```
run.complete.ss(
  n2, beta.A2,
  censoring.limit.2,
  true.estimates.strong,
  generate.plots = TRUE
)
```




MS−Cox RMST
RMST
0.0
−0.5
−1.0
0
1
2
3
4
Time
MS−Cox
Marginal



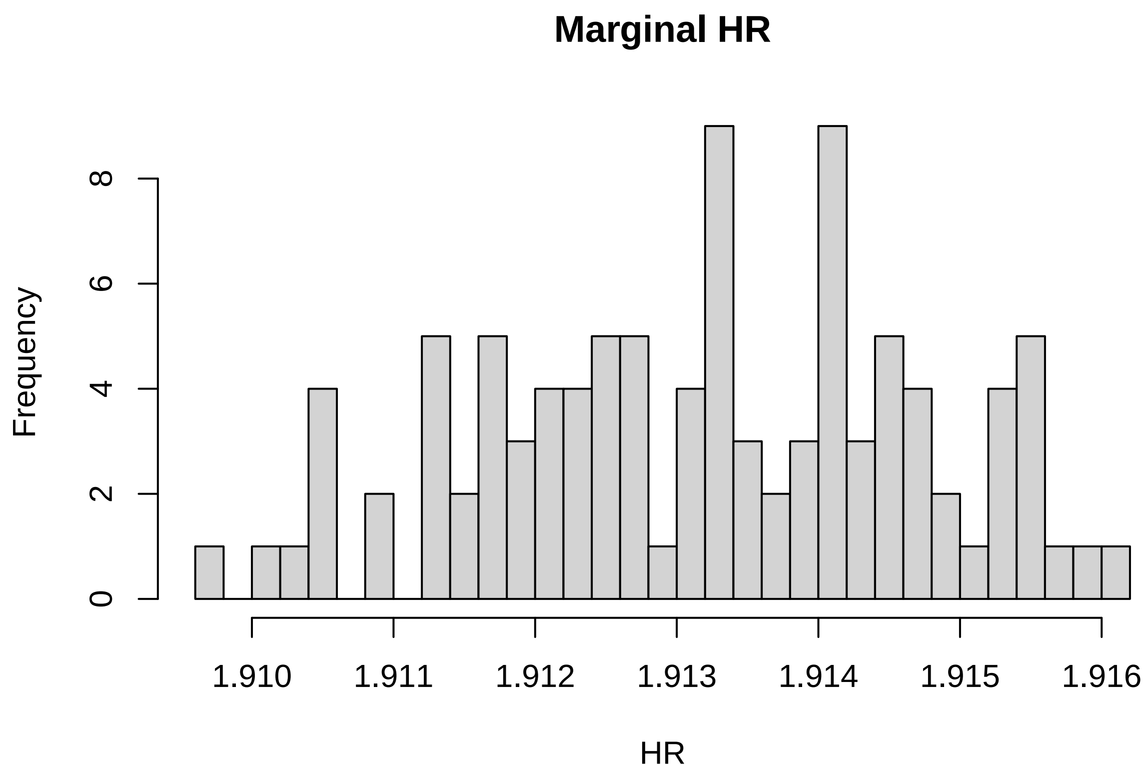
Case Base RMST
RMST
0.0
−0.5
−1.0
Case Base
Marginal
0
1
2
3
4
Time

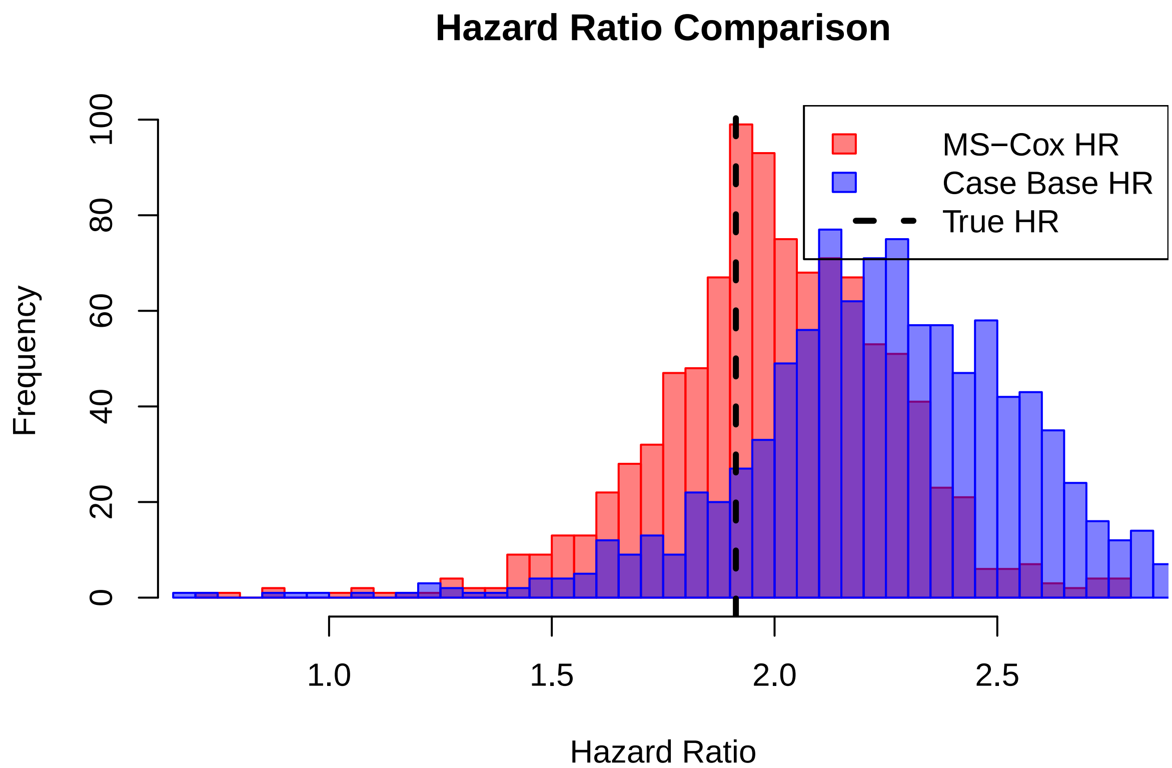


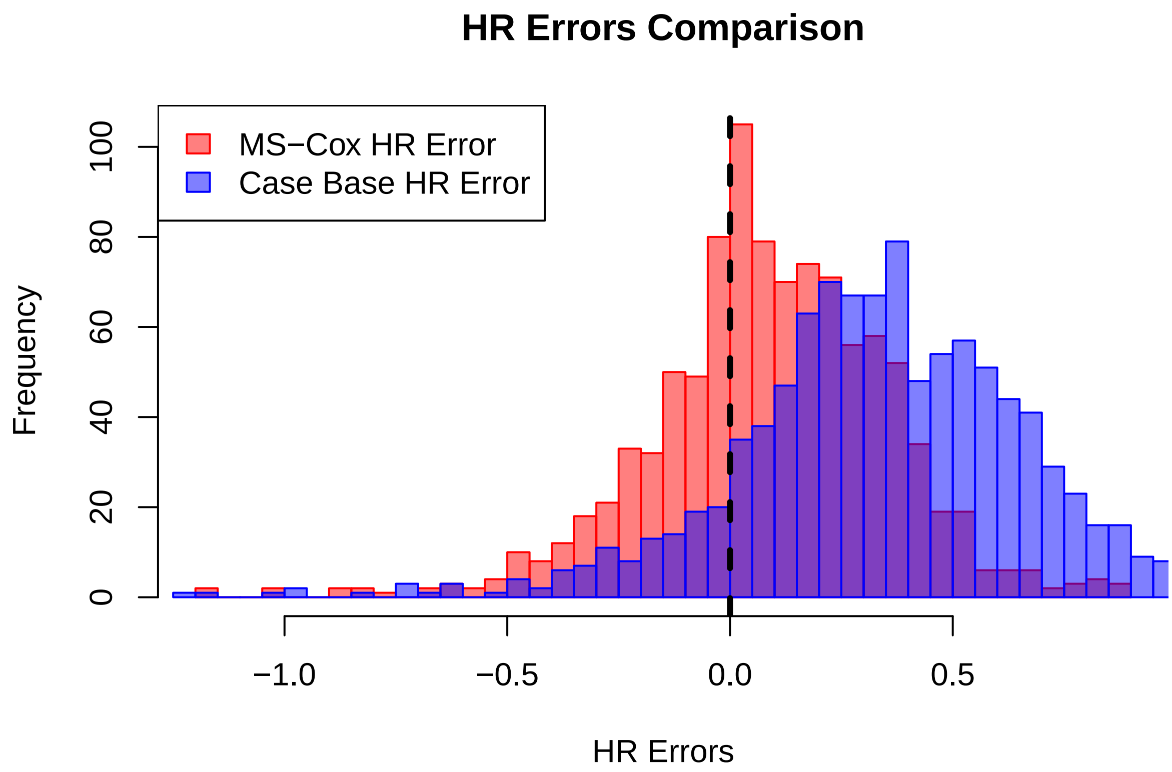

```
##
## ========================================
##
## True Mean Restricted Mean Survival Time
## ========================================
## 1 : -0.1676967
## 2 : -0.400371
## 3 : -0.6227171
## 4 : -0.8245784
##
##
## ========================================
## MS-Cox RMST
## ========================================
##
## Restricted Mean Survival Time Summary
## ----------------------------------------
##
## Tau = 1
## ----------------------------------------
## Mean RMST: -0.1737996
## Bias: -0.006102953  | MC-SE: 0.001097309
## RMSE: 0.03521546
## Mean Estimated SE: 0.04181334  | Empirical SD: 0.03469995
## Coverage: 0.985  | MC-SE: 0.003843826
##
```

```
## Tau = 2
## ----------------------------------------
## Mean RMST: -0.4176551
## Bias: -0.01728412  | MC-SE: 0.00268541
## RMSE: 0.08661959
## Mean Estimated SE: 0.09088202  | Empirical SD: 0.08492011
## Coverage: 0.967  | MC-SE: 0.005648982
##
## Tau = 3
## ----------------------------------------
## Mean RMST: -0.6506802
## Bias: -0.02796312  | MC-SE: 0.0041905
## RMSE: 0.1353686
## Mean Estimated SE: 0.1316875  | Empirical SD: 0.1325153
## Coverage: 0.946  | MC-SE: 0.007147307
##
## Tau = 4
## ----------------------------------------
## Mean RMST: -0.8612529
## Bias: -0.03667449  | MC-SE: 0.005515439
## RMSE: 0.1781423
## Mean Estimated SE: 0.1646598  | Empirical SD: 0.1744135
## Coverage: 0.939  | MC-SE: 0.007568289
##
##
## ========================================
## Case Base RMST
## ========================================
##
## Restricted Mean Survival Time Summary
## ----------------------------------------
##
## Tau = 1
## ----------------------------------------
## Mean RMST: -0.1662189
## Bias: 0.001477707  | MC-SE: 0.001101763
## RMSE: 0.03485473
## Mean Estimated SE: 0.02469436  | Empirical SD: 0.03484082
## Coverage: 0.879  | MC-SE: 0.01031305
##
## Tau = 2
## ----------------------------------------
## Mean RMST: -0.4464731
## Bias: -0.0461021  | MC-SE: 0.002834081
## RMSE: 0.1007442
## Mean Estimated SE: 0.06902601  | Empirical SD: 0.08962151
## Coverage: 0.794  | MC-SE: 0.01278921
##
## Tau = 3
## ----------------------------------------
## Mean RMST: -0.6983467
## Bias: -0.07562958  | MC-SE: 0.004364929
## RMSE: 0.1573321
## Mean Estimated SE: 0.1139078  | Empirical SD: 0.1380312
```

```
## Coverage: 0.815  | MC-SE: 0.01227905
##
## Tau = 4
## ----------------------------------------
## Mean RMST: -0.8913959
## Bias: -0.06681749  | MC-SE: 0.005584178
## RMSE: 0.1887232
## Mean Estimated SE: 0.1535955  | Empirical SD: 0.1765872
## Coverage: 0.895  | MC-SE: 0.00969407
```




Hazard Ratio Comparison
MS−Cox HR
Case Base HR
True HR
Frequency
0
20
40
60
80
100
1.0
1.5
2.0
2.5
Hazard Ratio

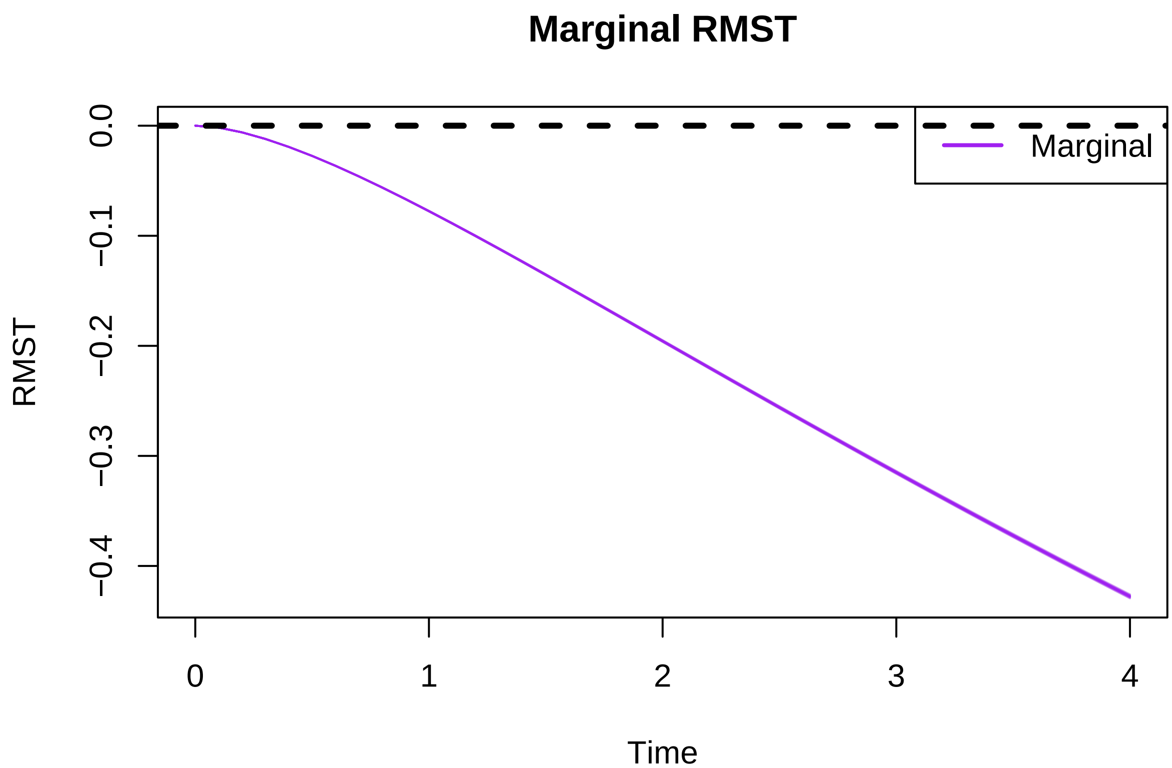


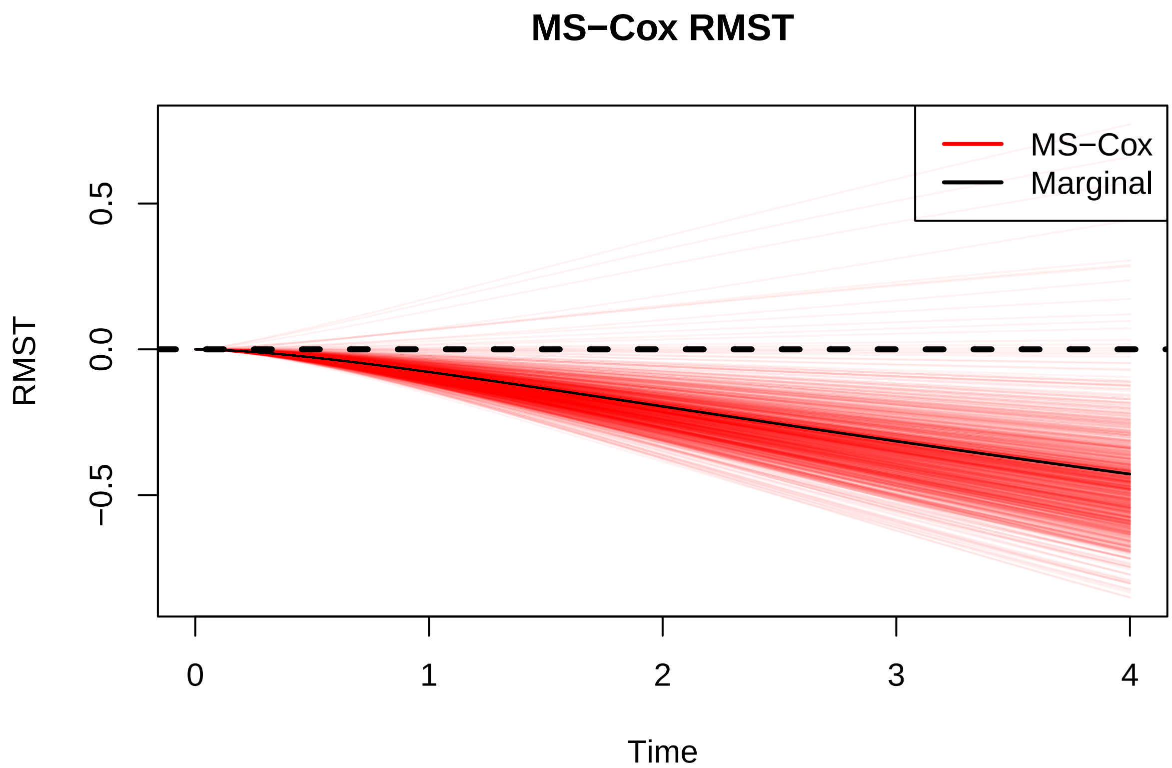

```
##
## ========================================
## True HR: 1.913181
## ========================================
##
## ========================================
## MS-Cox
## ========================================
##
## Hazard Ratio Summary
## ----------------------------------------
## Mean HR: 1.983243
## Bias: 0.07006206  | MC-SE: 0.008500306
## MSE: 0.08026432
## RMSE: 0.2833096
## Mean Estimated SE: 0.1417184  | Empirical SD: 0.2688033
## Type I Error/Power: 0.96
## Coverage: 0.945  | MC-SE: 0.007209369
##
##
## ========================================
## Case Base
## ========================================
##
## Hazard Ratio Summary
## ----------------------------------------
```

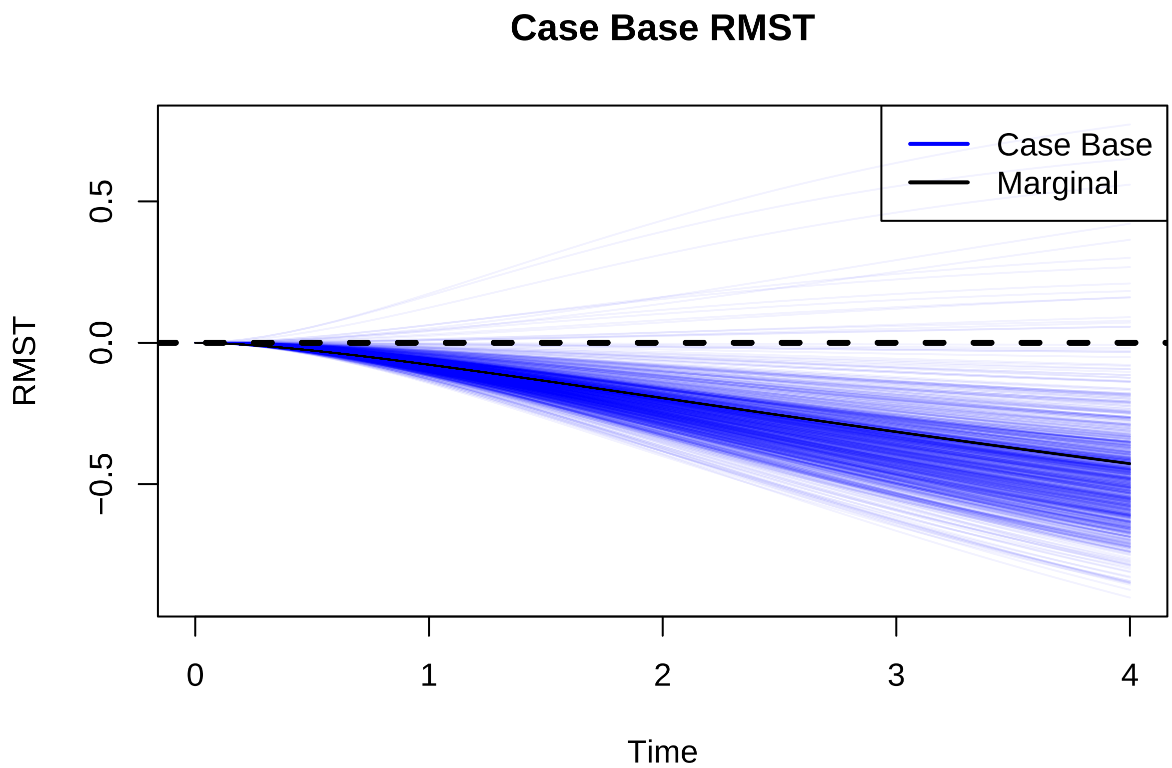
```
## Mean HR: 2.228819
## Bias: 0.3156383  | MC-SE: 0.01050939
## MSE: 0.2279821
## RMSE: 0.4774747
## Mean Estimated SE: 0.134544  | Empirical SD: 0.3323361
## Type I Error/Power: 0.978
## Coverage: 0.69  | MC-SE: 0.01462532
```

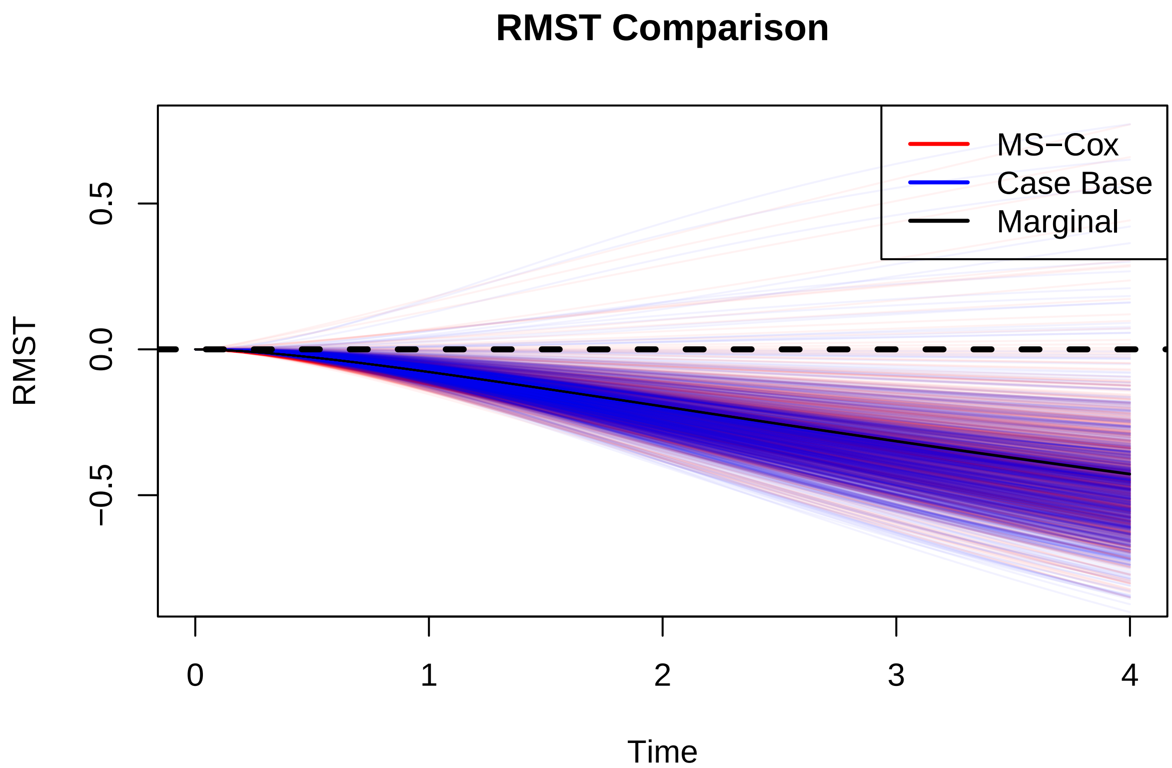
Run Simulation Study: Setting 3

```
run.complete.ss(
  n3, beta.A3,
  censoring.limit.3,
  true.estimates.moderate,
  generate.plots = TRUE
)
```




MS−Cox RMST
MS−Cox
Marginal
RMST
0.5
0.0
−0.5
0
1
2
3
4
Time

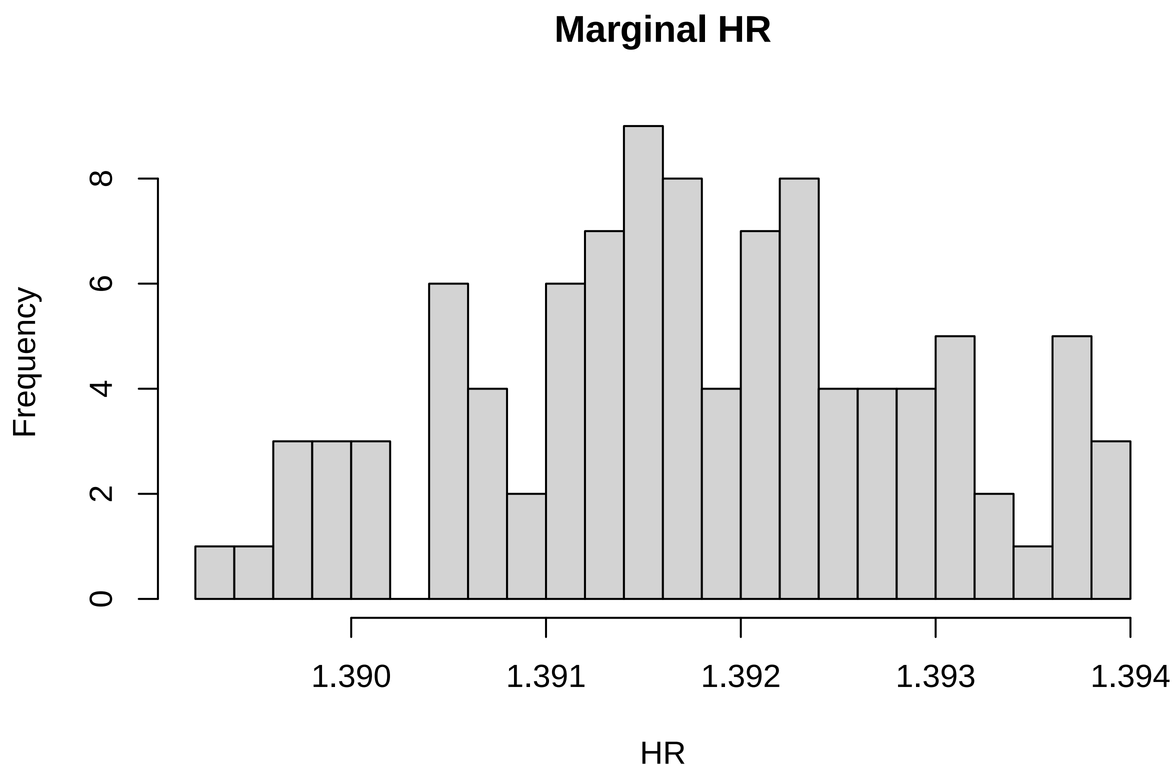

Case Base RMST
Case Base
Marginal
RMST
0.5
0.0
−0.5
0
1
2
3
4
Time

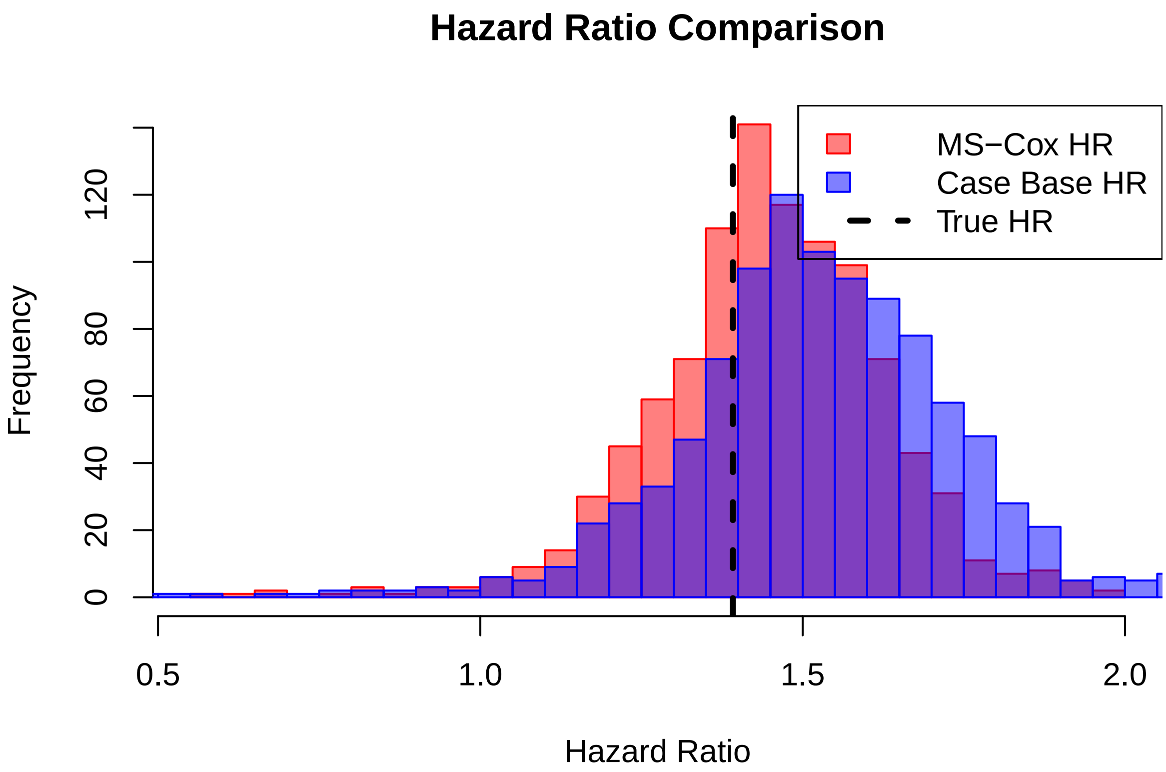


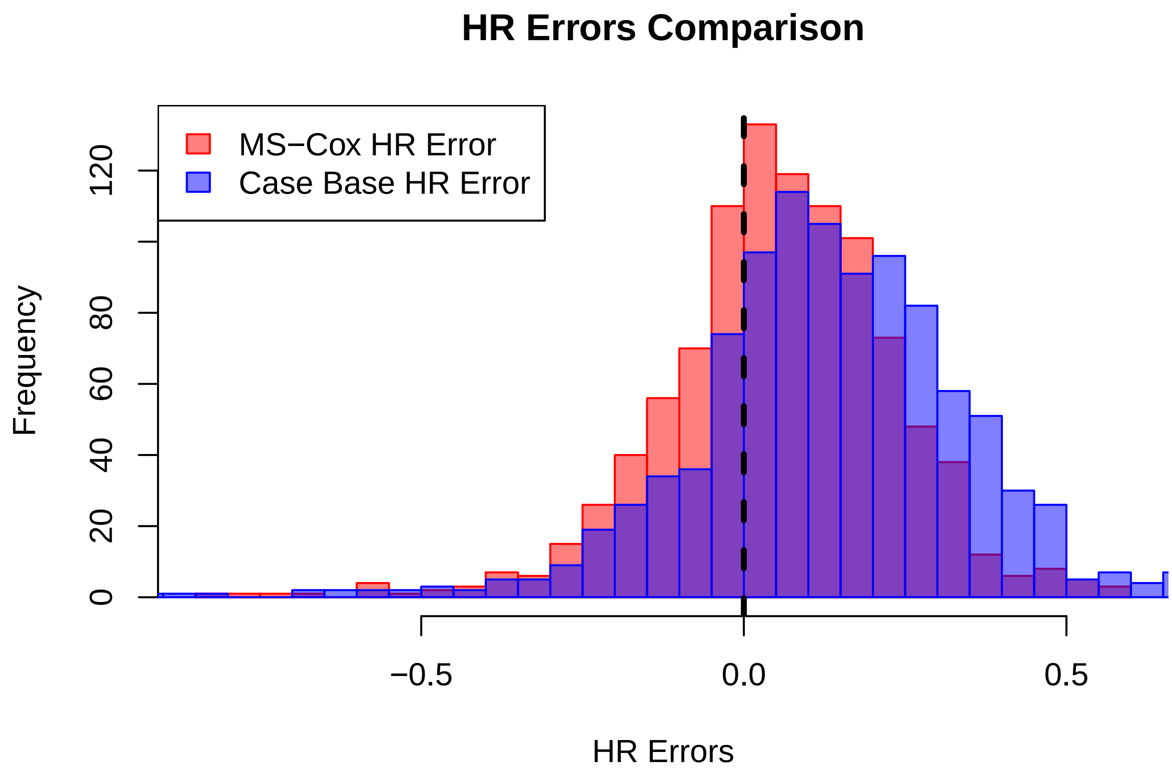

```
##
## ========================================
##
## True Mean Restricted Mean Survival Time
## ========================================
## 1 : -0.07760737
## 2 : -0.1956718
## 3 : -0.3150435
## 4 : -0.4276574
##
##
## ========================================
## MS-Cox RMST
## ========================================
##
## Restricted Mean Survival Time Summary
## ----------------------------------------
##
## Tau = 1
## ----------------------------------------
## Mean RMST: -0.08267714
## Bias: -0.005069766  | MC-SE: 0.0009690469
## RMSE: 0.03104538
## Mean Estimated SE: 0.03463114  | Empirical SD: 0.03064395
## Coverage: 0.982  | MC-SE: 0.004204284
##
```

```
## Tau = 2
## ----------------------------------------
## Mean RMST: -0.2108824
## Bias: -0.01521067  | MC-SE: 0.002454862
## RMSE: 0.07906759
## Mean Estimated SE: 0.08376786  | Empirical SD: 0.07762954
## Coverage: 0.969  | MC-SE: 0.005480785
##
## Tau = 3
## ----------------------------------------
## Mean RMST: -0.3409267
## Bias: -0.0258832  | MC-SE: 0.003959439
## RMSE: 0.1277944
## Mean Estimated SE: 0.1306268  | Empirical SD: 0.1252085
## Coverage: 0.966  | MC-SE: 0.005730969
##
## Tau = 4
## ----------------------------------------
## Mean RMST: -0.4633661
## Bias: -0.03570873  | MC-SE: 0.005368086
## RMSE: 0.1733858
## Mean Estimated SE: 0.1726648  | Empirical SD: 0.1697538
## Coverage: 0.961  | MC-SE: 0.006122009
##
##
## ========================================
## Case Base RMST
## ========================================
##
## Restricted Mean Survival Time Summary
## ----------------------------------------
##
## Tau = 1
## ----------------------------------------
## Mean RMST: -0.07468256
## Bias: 0.002924818  | MC-SE: 0.000914879
## RMSE: 0.02906409
## Mean Estimated SE: 0.02342581  | Empirical SD: 0.02893101
## Coverage: 0.931  | MC-SE: 0.008014924
##
## Tau = 2
## ----------------------------------------
## Mean RMST: -0.2164729
## Bias: -0.02080118  | MC-SE: 0.002574241
## RMSE: 0.08398084
## Mean Estimated SE: 0.06868738  | Empirical SD: 0.08140466
## Coverage: 0.902  | MC-SE: 0.009401915
##
## Tau = 3
## ----------------------------------------
## Mean RMST: -0.3593907
## Bias: -0.04434726  | MC-SE: 0.004202496
## RMSE: 0.1400357
## Mean Estimated SE: 0.1159593  | Empirical SD: 0.1328946
```

```
## Coverage: 0.9  | MC-SE: 0.009486833
##
## Tau = 4
## ----------------------------------------
## Mean RMST: -0.4802259
## Bias: -0.0525685  | MC-SE: 0.005571007
## RMSE: 0.1837621
## Mean Estimated SE: 0.1578835  | Empirical SD: 0.1761707
## Coverage: 0.915  | MC-SE: 0.008819014
```




Hazard Ratio Comparison
MS−Cox HR
Case Base HR
True HR
Frequency
0
20
40
60
80
120
0.5
1.0
1.5
2.0
Hazard Ratio

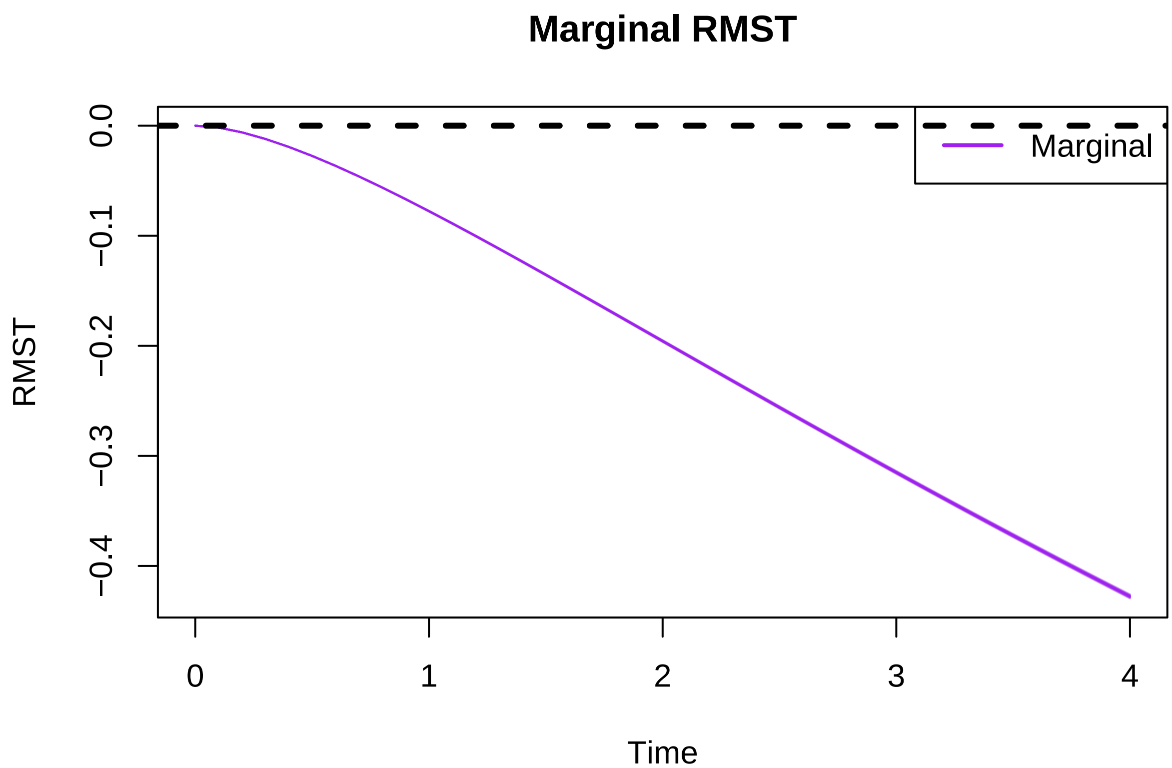


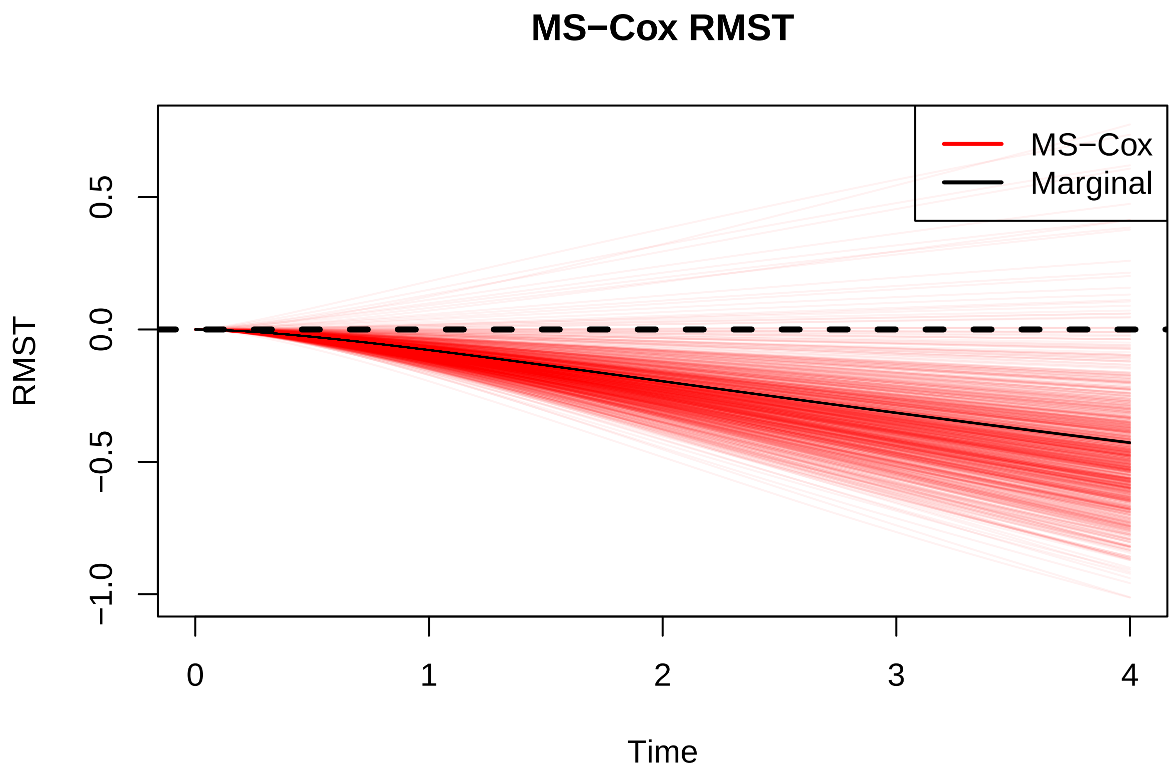

```
##
## ========================================
## True HR: 1.391778
## ========================================
##
## ========================================
## MS-Cox
## ========================================
##
## Hazard Ratio Summary
## ----------------------------------------
## Mean HR: 1.435647
## Bias: 0.04386899  | MC-SE: 0.005693821
## MSE: 0.03553542
## RMSE: 0.1885084
## Mean Estimated SE: 0.1354279  | Empirical SD: 0.1800544
## Type I Error/Power: 0.789
## Coverage: 0.959  | MC-SE: 0.006270486
##
##
## ========================================
## Case Base
## ========================================
##
## Hazard Ratio Summary
## ----------------------------------------
```

```
## Mean HR: 1.504976
## Bias: 0.1131975  | MC-SE: 0.006662953
## MSE: 0.06112684
## RMSE: 0.2472384
## Mean Estimated SE: 0.135057  | Empirical SD: 0.2107011
## Type I Error/Power: 0.85
## Coverage: 0.869  | MC-SE: 0.01066954
```

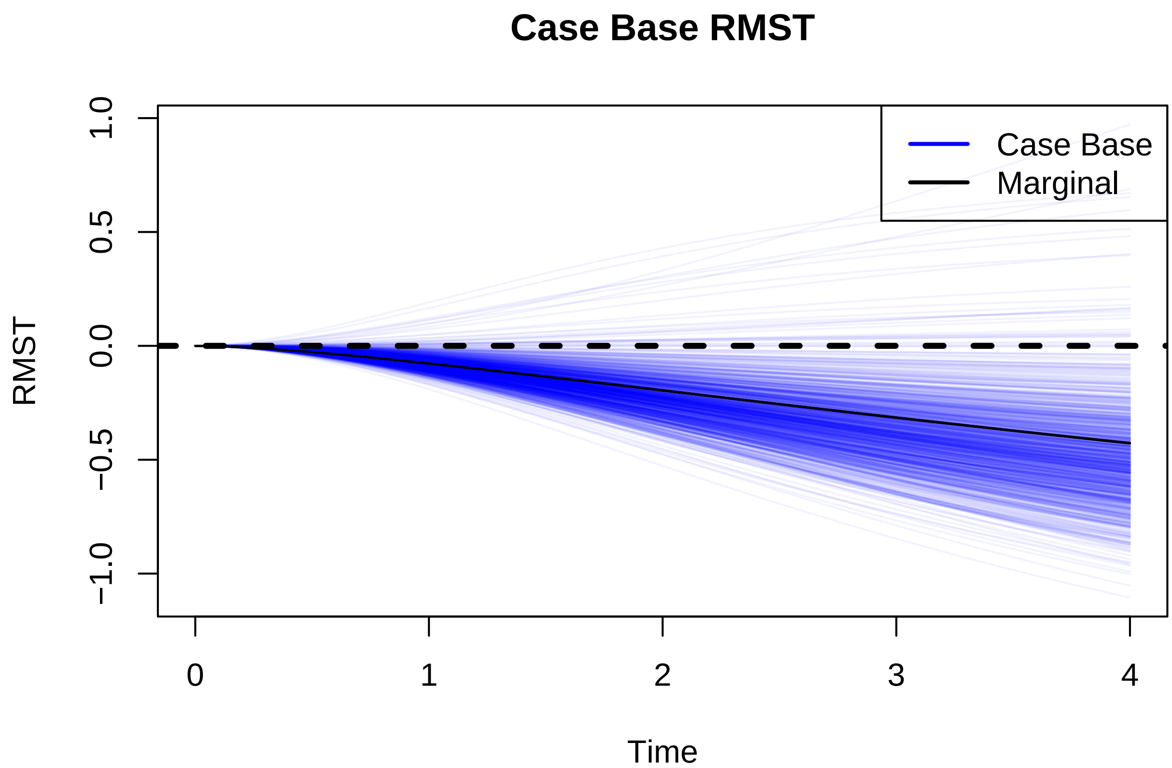
Run Simulation Study: Setting 4

```
run.complete.ss(
  n4, beta.A4,
  censoring.limit.4,
  true.estimates.moderate,
  generate.plots = TRUE
)
```

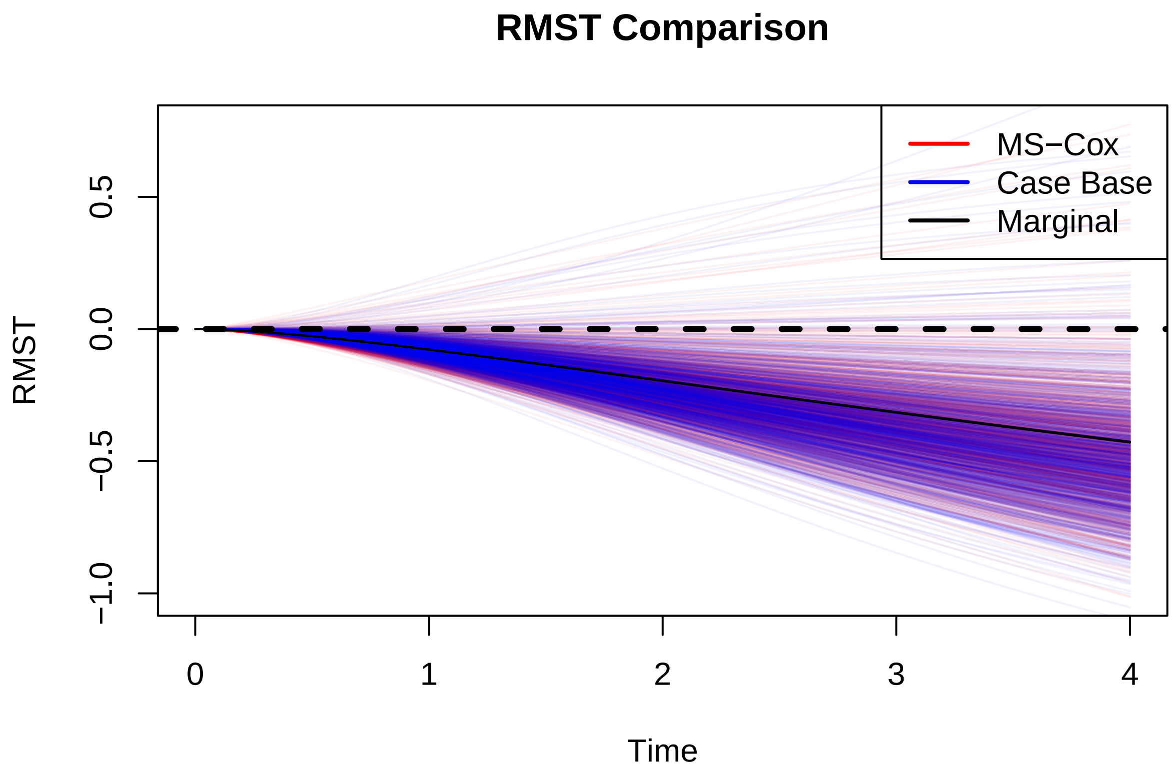




MS−Cox RMST
MS−Cox
Marginal
RMST
0.5
0.0
−0.5
−1.0
0
1
2
3
4
Time



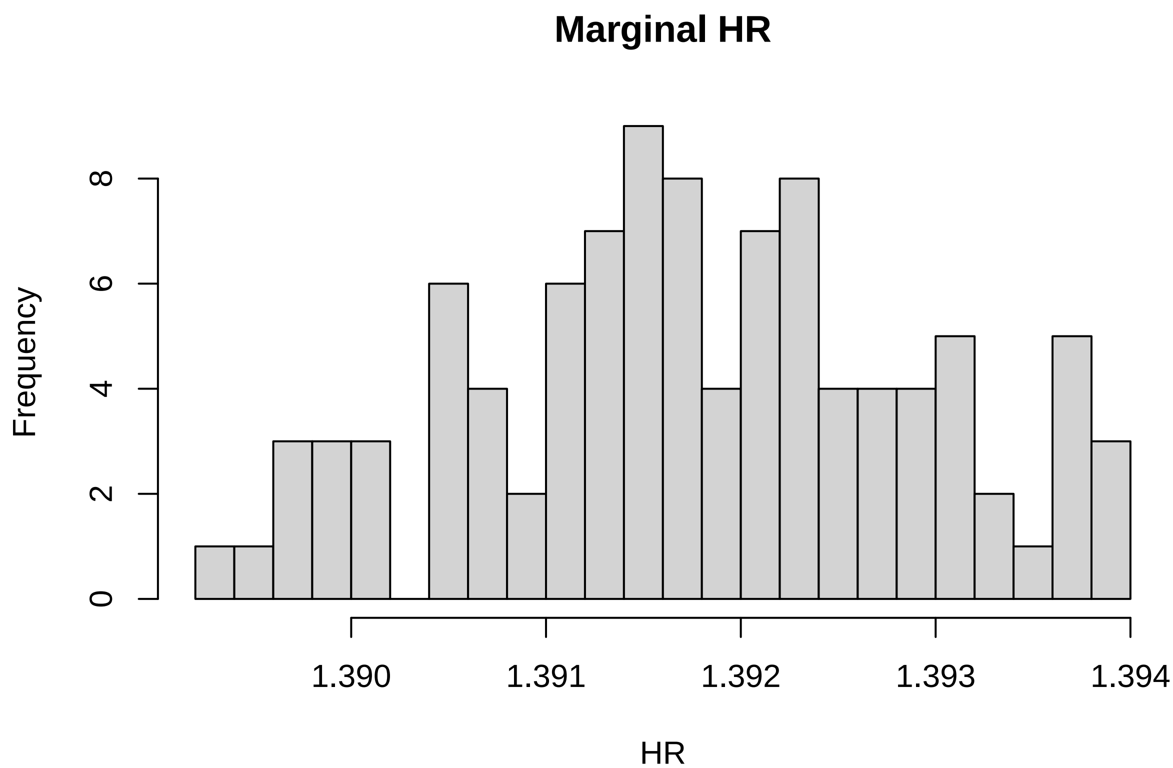
Case Base RMST
Case Base
Marginal
RMST
Time
1.0
0.5
0.0
−0.5
−1.0
0
1
2
3
4

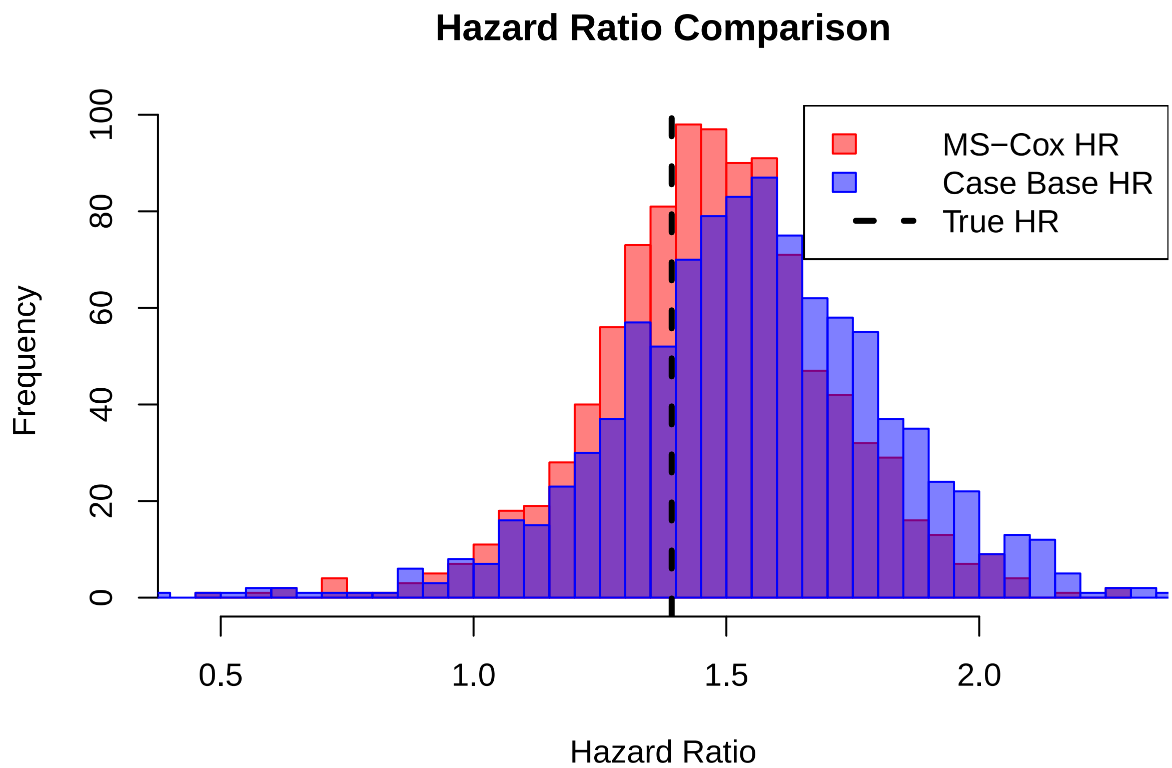



```
##
## ========================================
##
## True Mean Restricted Mean Survival Time
## ========================================
## 1 : -0.07760737
## 2 : -0.1956718
## 3 : -0.3150435
## 4 : -0.4276574
##
##
## ========================================
## MS-Cox RMST
## ========================================
##
## Restricted Mean Survival Time Summary
## ----------------------------------------
##
## Tau = 1
## ----------------------------------------
## Mean RMST: -0.08507421
## Bias: -0.007466837  | MC-SE: 0.001219094
## RMSE: 0.03924867
## Mean Estimated SE: 0.04599072  | Empirical SD: 0.03855114
## Coverage: 0.988  | MC-SE: 0.003443254
##
```

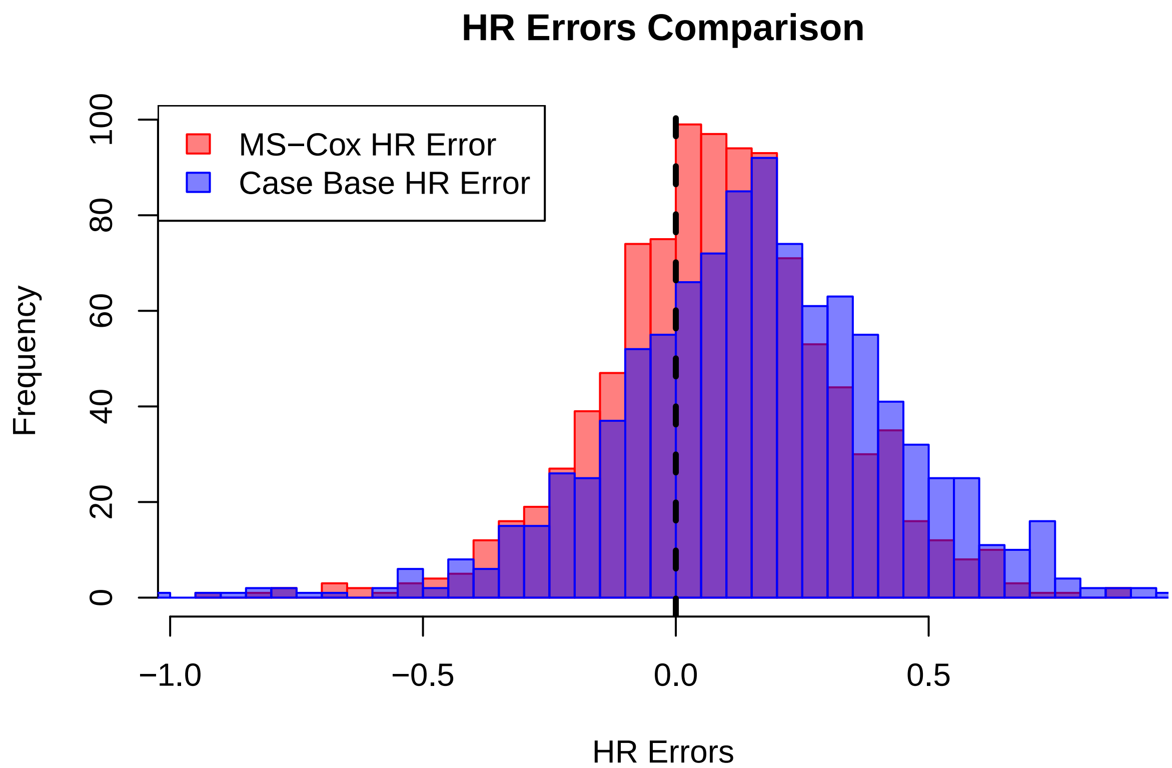
```
## Tau = 2
## ----------------------------------------
## Mean RMST: -0.2178636
## Bias: -0.02219185  | MC-SE: 0.003083781
## RMSE: 0.09996336
## Mean Estimated SE: 0.1111314  | Empirical SD: 0.09751771
## Coverage: 0.974  | MC-SE: 0.005032296
##
## Tau = 3
## ----------------------------------------
## Mean RMST: -0.3526093
## Bias: -0.03756587  | MC-SE: 0.004973146
## RMSE: 0.1616126
## Mean Estimated SE: 0.1730591  | Empirical SD: 0.1572647
## Coverage: 0.966  | MC-SE: 0.005730969
##
## Tau = 4
## ----------------------------------------
## Mean RMST: -0.4793477
## Bias: -0.05169031  | MC-SE: 0.006745406
## RMSE: 0.2193784
## Mean Estimated SE: 0.2284912  | Empirical SD: 0.2133085
## Coverage: 0.955  | MC-SE: 0.006555532
##
##
## ========================================
## Case Base RMST
## ========================================
##
## Restricted Mean Survival Time Summary
## ----------------------------------------
##
## Tau = 1
## ----------------------------------------
## Mean RMST: -0.07677088
## Bias: 0.0008364894  | MC-SE: 0.001177639
## RMSE: 0.03723098
## Mean Estimated SE: 0.03191703  | Empirical SD: 0.03724021
## Coverage: 0.933  | MC-SE: 0.00790639
##
## Tau = 2
## ----------------------------------------
## Mean RMST: -0.2222491
## Bias: -0.02657733  | MC-SE: 0.003301822
## RMSE: 0.1076916
## Mean Estimated SE: 0.09294632  | Empirical SD: 0.1044128
## Coverage: 0.914  | MC-SE: 0.00886589
##
## Tau = 3
## ----------------------------------------
## Mean RMST: -0.3688103
## Bias: -0.05376684  | MC-SE: 0.005390803
## RMSE: 0.1786689
## Mean Estimated SE: 0.156323  | Empirical SD: 0.1704722
```

```
## Coverage: 0.91  | MC-SE: 0.009049862
##
## Tau = 4
## ---------------------------------------
## Mean RMST: -0.4928804
## Bias: -0.06522303  | MC-SE: 0.007164089
## RMSE: 0.2356414
## Mean Estimated SE: 0.2124701  | Empirical SD: 0.2265484
## Coverage: 0.927  | MC-SE: 0.008226239
```





Hazard Ratio Comparison
MS−Cox HR
Case Base HR
True HR
Frequency
0
20
40
60
80
100
0.5
1.0
1.5
2.0
Hazard Ratio

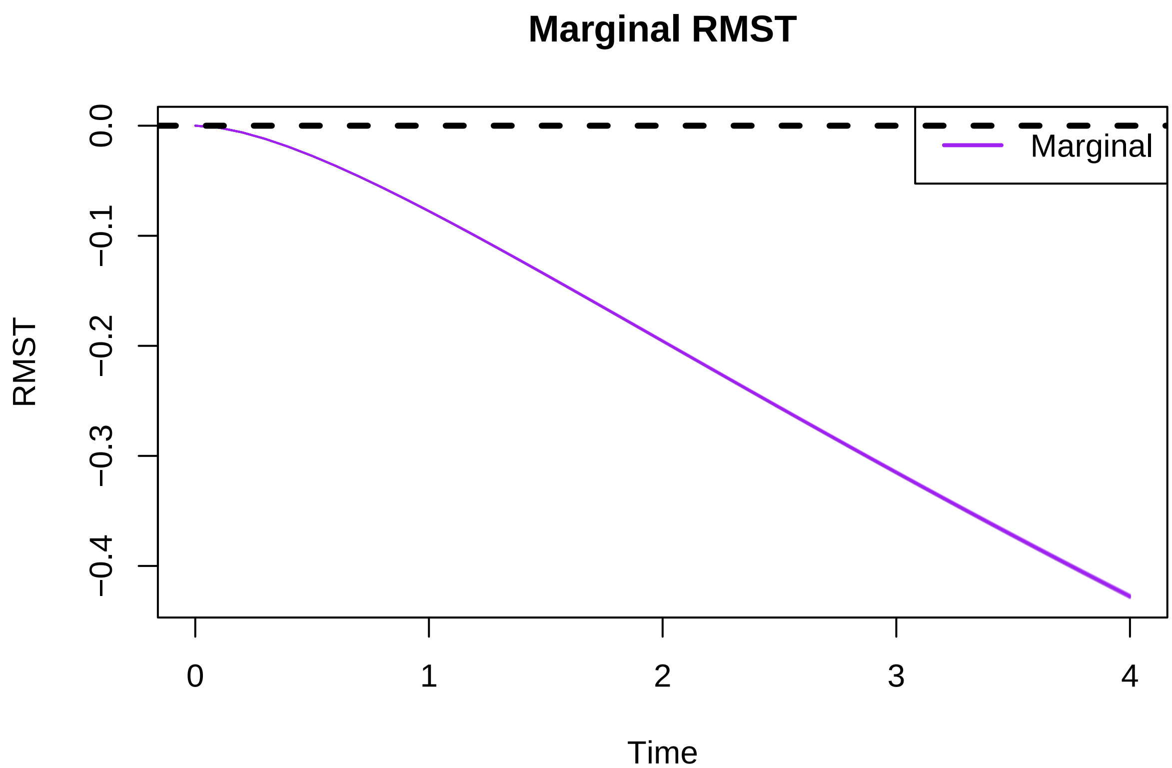


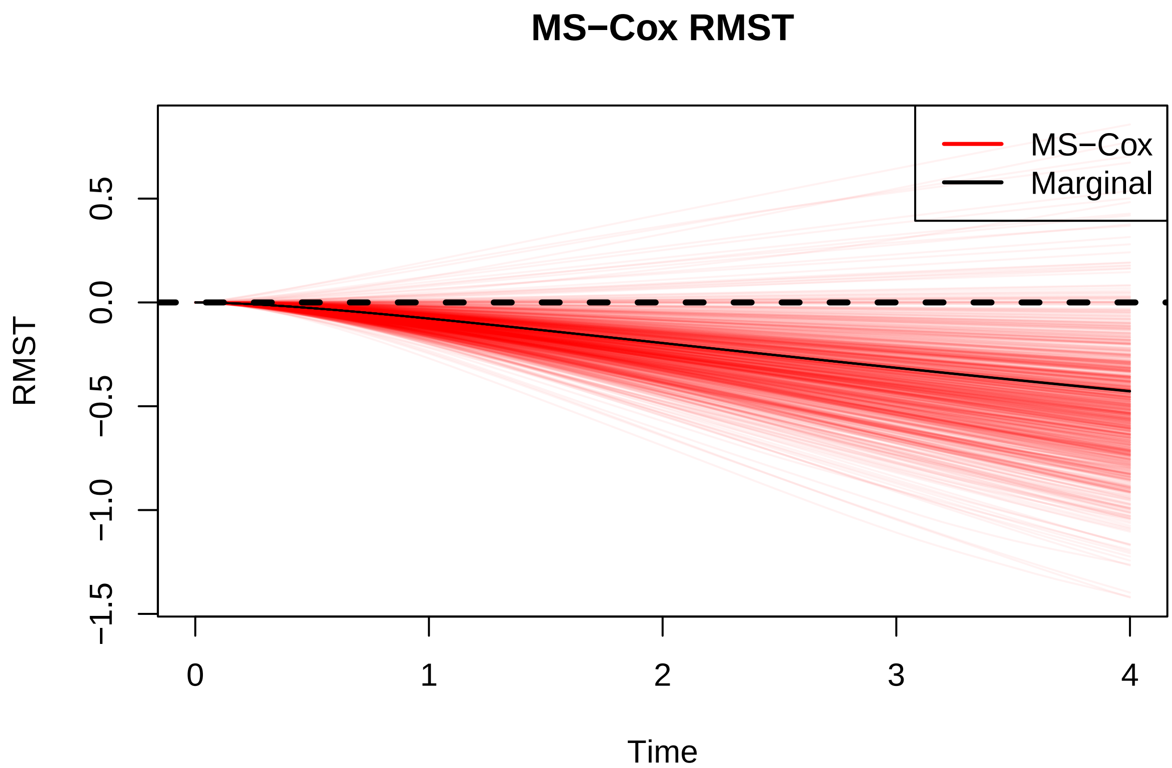

```
##
## ========================================
## True HR: 1.391778
## ========================================
##
## ========================================
## MS-Cox
## ========================================
##
## Hazard Ratio Summary
## ----------------------------------------
## Mean HR: 1.456399
## Bias: 0.06462114  | MC-SE: 0.007377941
## MSE: 0.06153404
## RMSE: 0.2480606
## Mean Estimated SE: 0.1802018  | Empirical SD: 0.233311
## Type I Error/Power: 0.639
## Coverage: 0.956  | MC-SE: 0.006485677
##
##
## ========================================
## Case Base
## ========================================
##
## Hazard Ratio Summary
## ----------------------------------------
```

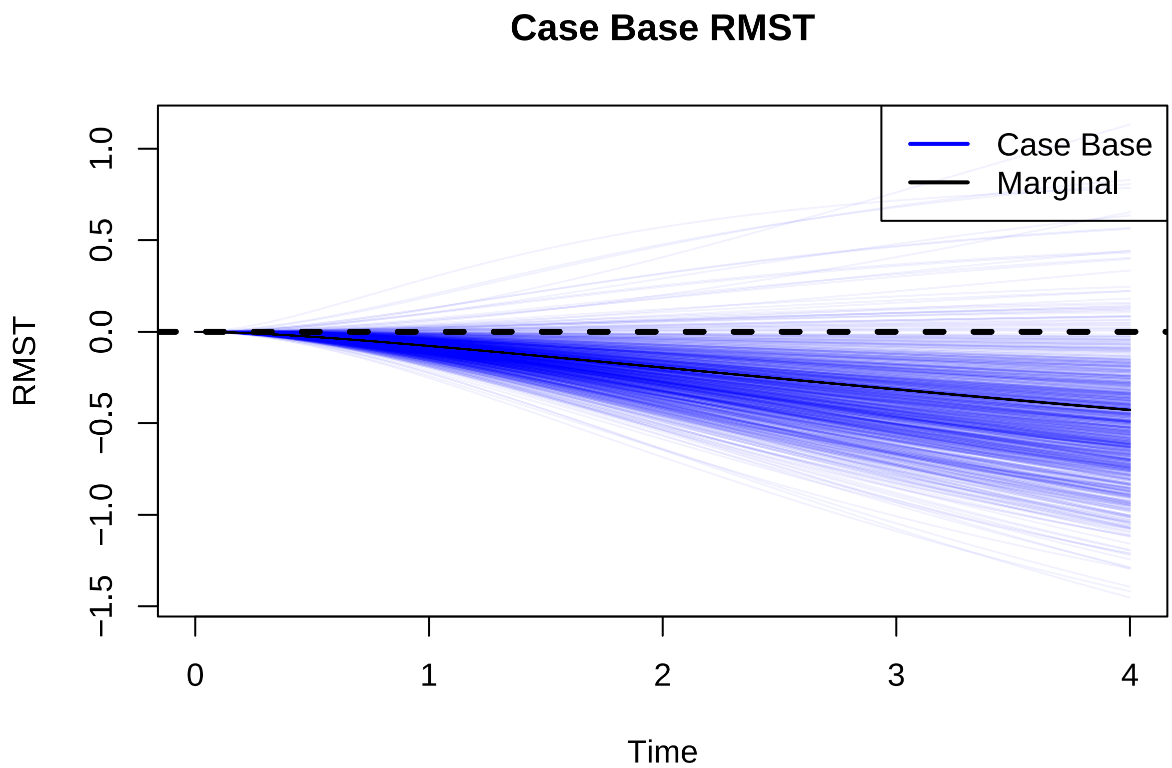
```
## Mean HR: 1.522871
## Bias: 0.1310925  | MC-SE: 0.008807012
## MSE: 0.1027226
## RMSE: 0.3205036
## Mean Estimated SE: 0.1829533  | Empirical SD: 0.2785022
## Type I Error/Power: 0.711
## Coverage: 0.903  | MC-SE: 0.009359006
```

Run Simulation Study: Setting 5

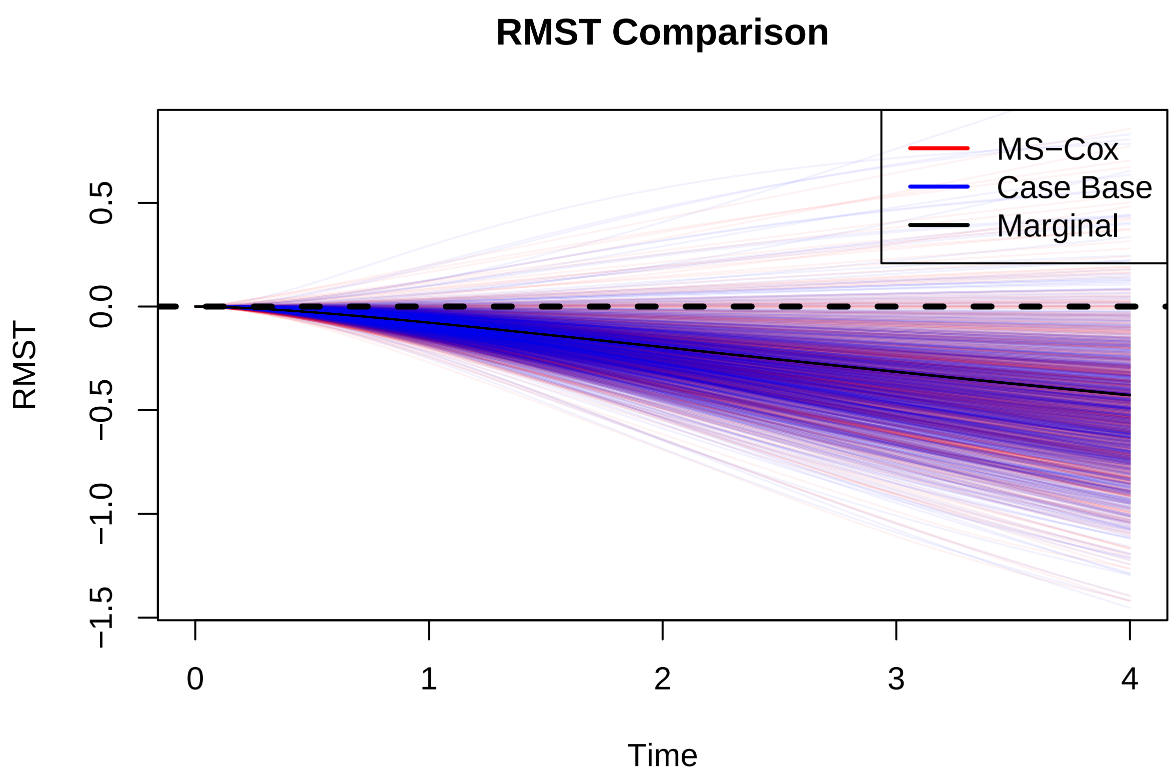
```r
run.complete.ss(
  n5, beta.A5,
  censoring.limit.5,
  true.estimates.moderate,
  generate.plots = TRUE
)
```




MS−Cox RMST
MS−Cox
Marginal
RMST
0.5
0.0
−0.5
−1.0
−1.5
0
1
2
3
4
Time



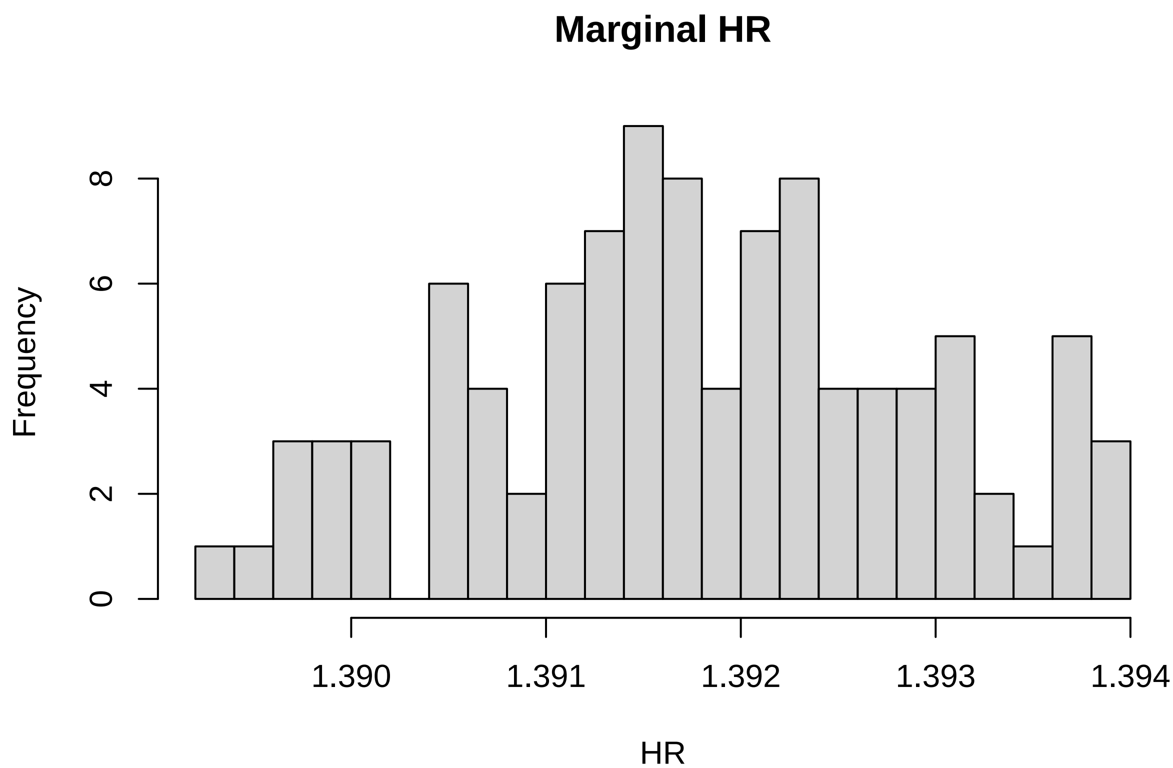
Case Base RMST
Case Base
Marginal
RMST
1.0
0.5
0.0
−0.5
−1.0
−1.5
0
1
2
3
4
Time

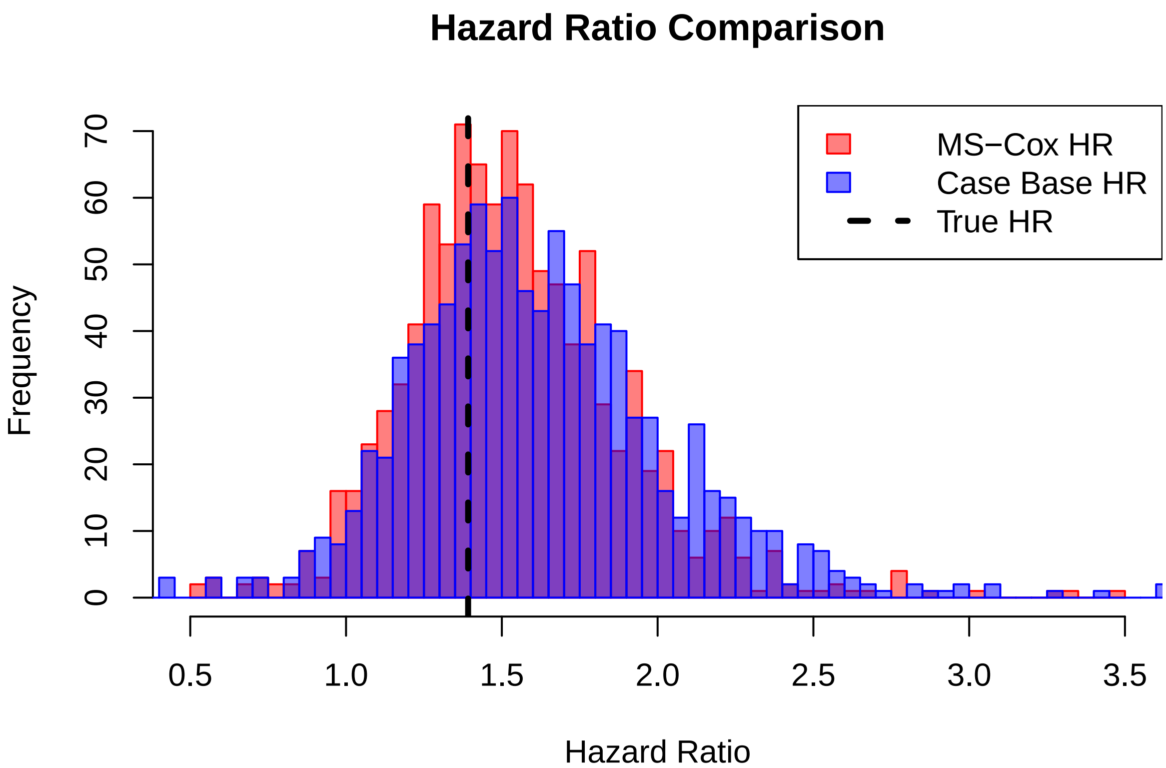


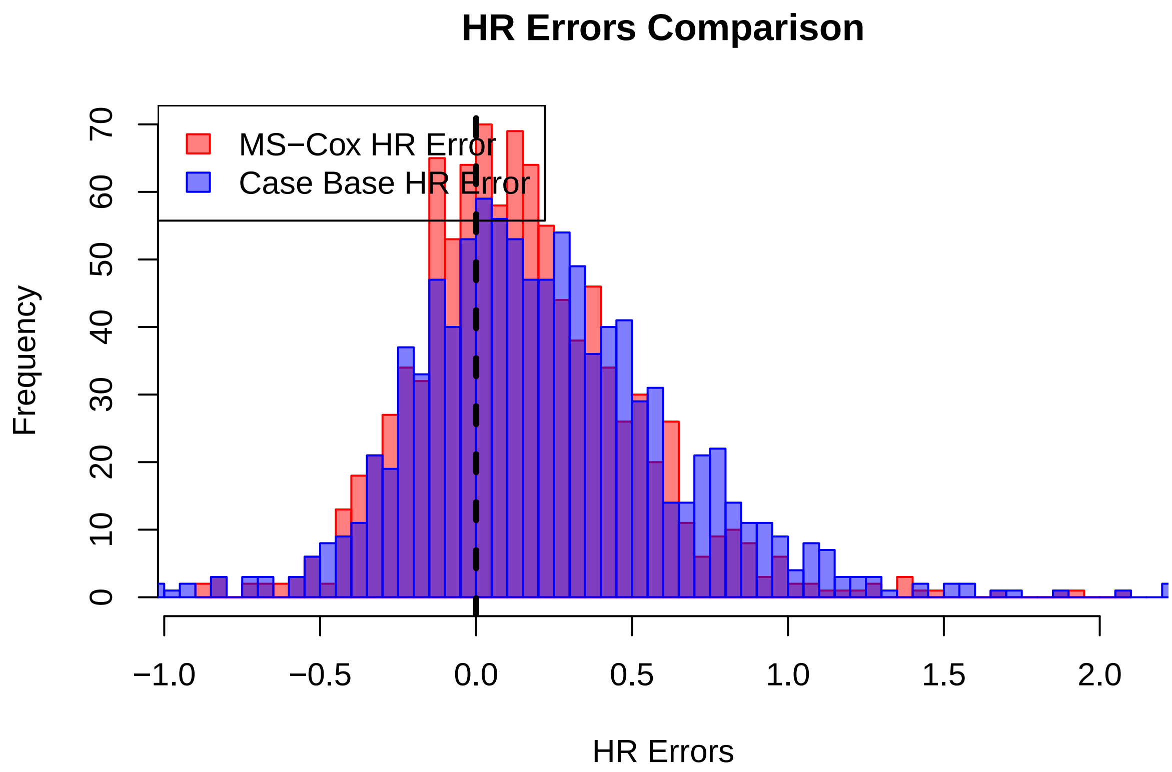

```
##
## ========================================
##
## True Mean Restricted Mean Survival Time
## ========================================
## 1 : -0.07760737
## 2 : -0.1956718
## 3 : -0.3150435
## 4 : -0.4276574
##
##
## ========================================
## MS-Cox RMST
## ========================================
##
## Restricted Mean Survival Time Summary
## ----------------------------------------
##
## Tau = 1
## ----------------------------------------
## Mean RMST: -0.09024178
## Bias: -0.01263441  | MC-SE: 0.00161456
## RMSE: 0.0525721
## Mean Estimated SE: 0.06070955  | Empirical SD: 0.05105687
## Coverage: 0.981  | MC-SE: 0.004317291
##
```

```
## Tau = 2
## ----------------------------------------
## Mean RMST: -0.2324634
## Bias: -0.0367916  | MC-SE: 0.004121714
## RMSE: 0.1353704
## Mean Estimated SE: 0.1466551  | Empirical SD: 0.13034
## Coverage: 0.97  | MC-SE: 0.005394442
##
## Tau = 3
## ----------------------------------------
## Mean RMST: -0.3768026
## Bias: -0.06175909  | MC-SE: 0.006669824
## RMSE: 0.219673
## Mean Estimated SE: 0.2277717  | Empirical SD: 0.2109183
## Coverage: 0.963  | MC-SE: 0.005969171
##
## Tau = 4
## ----------------------------------------
## Mean RMST: -0.5120562
## Bias: -0.08439885  | MC-SE: 0.009045422
## RMSE: 0.2980956
## Mean Estimated SE: 0.300006  | Empirical SD: 0.2860414
## Coverage: 0.946  | MC-SE: 0.007147307
##
##
## ========================================
## Case Base RMST
## ========================================
##
## Restricted Mean Survival Time Summary
## ----------------------------------------
##
## Tau = 1
## ----------------------------------------
## Mean RMST: -0.08120692
## Bias: -0.003599548  | MC-SE: 0.001594436
## RMSE: 0.05052367
## Mean Estimated SE: 0.04329993  | Empirical SD: 0.0504205
## Coverage: 0.923  | MC-SE: 0.008430362
##
## Tau = 2
## ----------------------------------------
## Mean RMST: -0.2345542
## Bias: -0.03888245  | MC-SE: 0.004435665
## RMSE: 0.1454898
## Mean Estimated SE: 0.1256959  | Empirical SD: 0.140268
## Coverage: 0.904  | MC-SE: 0.009315793
##
## Tau = 3
## ----------------------------------------
## Mean RMST: -0.3889512
## Bias: -0.07390769  | MC-SE: 0.007236111
## RMSE: 0.2403566
## Mean Estimated SE: 0.2113496  | Empirical SD: 0.2288259
```

```
## Coverage: 0.906  | MC-SE: 0.009228434
##
## Tau = 4
## ----------------------------------------
## Mean RMST: -0.5200658
## Bias: -0.09240845  | MC-SE: 0.009637085
## RMSE: 0.3183078
## Mean Estimated SE: 0.2876424  | Empirical SD: 0.3047514
## Coverage: 0.913  | MC-SE: 0.008912407
```




Hazard Ratio Comparison
MS−Cox HR
Case Base HR
True HR
Frequency
0
10
20
30
40
50
60
70
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Hazard Ratio

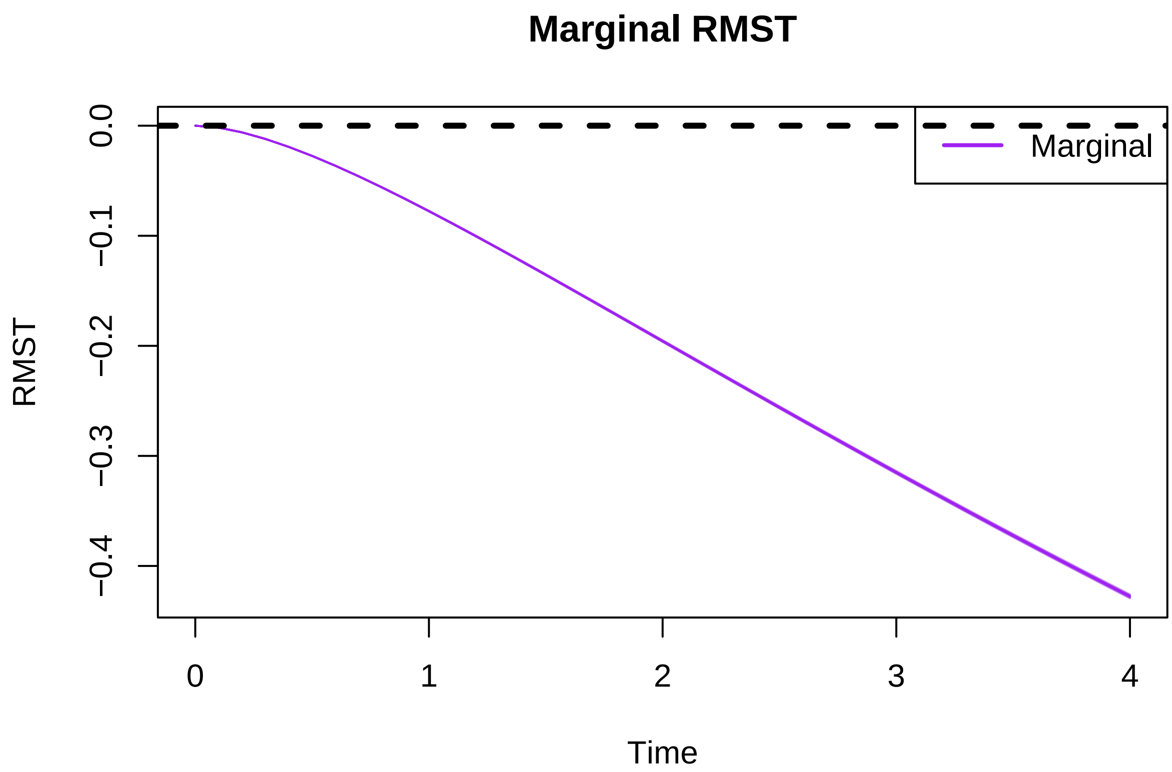


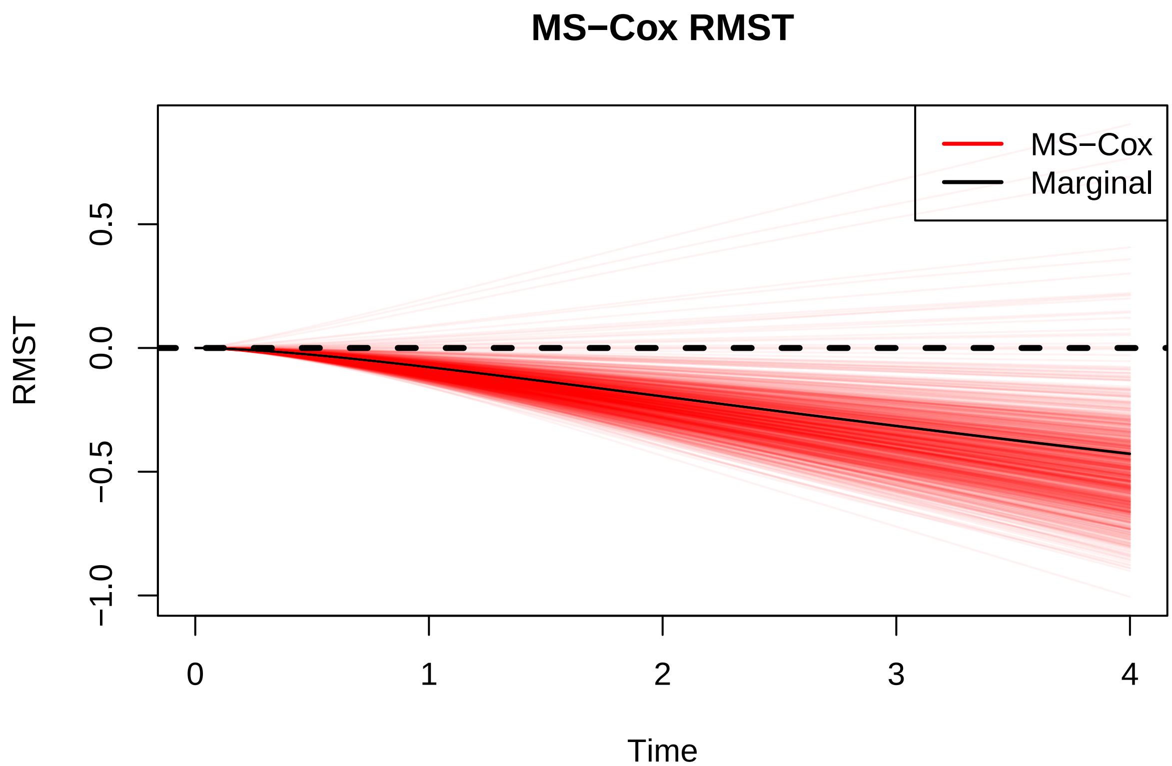

```
##
## ========================================
## True HR: 1.391778
## ========================================
##
## ========================================
## MS-Cox
## ========================================
##
## Hazard Ratio Summary
## ----------------------------------------
## Mean HR: 1.501318
## Bias: 0.1095401  | MC-SE: 0.01128092
## MSE: 0.1495646
## RMSE: 0.3867359
## Mean Estimated SE: 0.2392413  | Empirical SD: 0.356734
## Type I Error/Power: 0.441
## Coverage: 0.948  | MC-SE: 0.007021111
##
##
## ========================================
## Case Base
## ========================================
##
## Hazard Ratio Summary
## ----------------------------------------
```

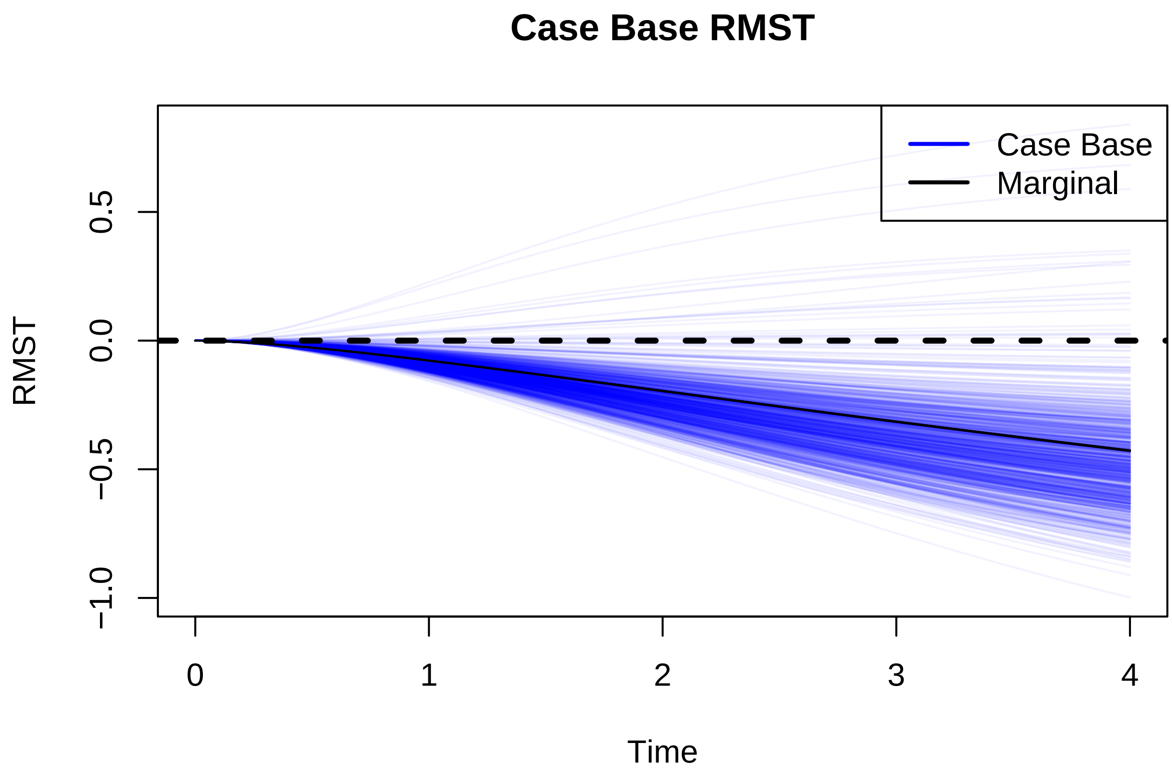
```
## Mean HR: 1.56233
## Bias: 0.170552  | MC-SE: 0.01322492
## MSE: 0.2254961
## RMSE: 0.4748643
## Mean Estimated SE: 0.2493552  | Empirical SD: 0.4182087
## Type I Error/Power: 0.495
## Coverage: 0.9  | MC-SE: 0.009486833
```

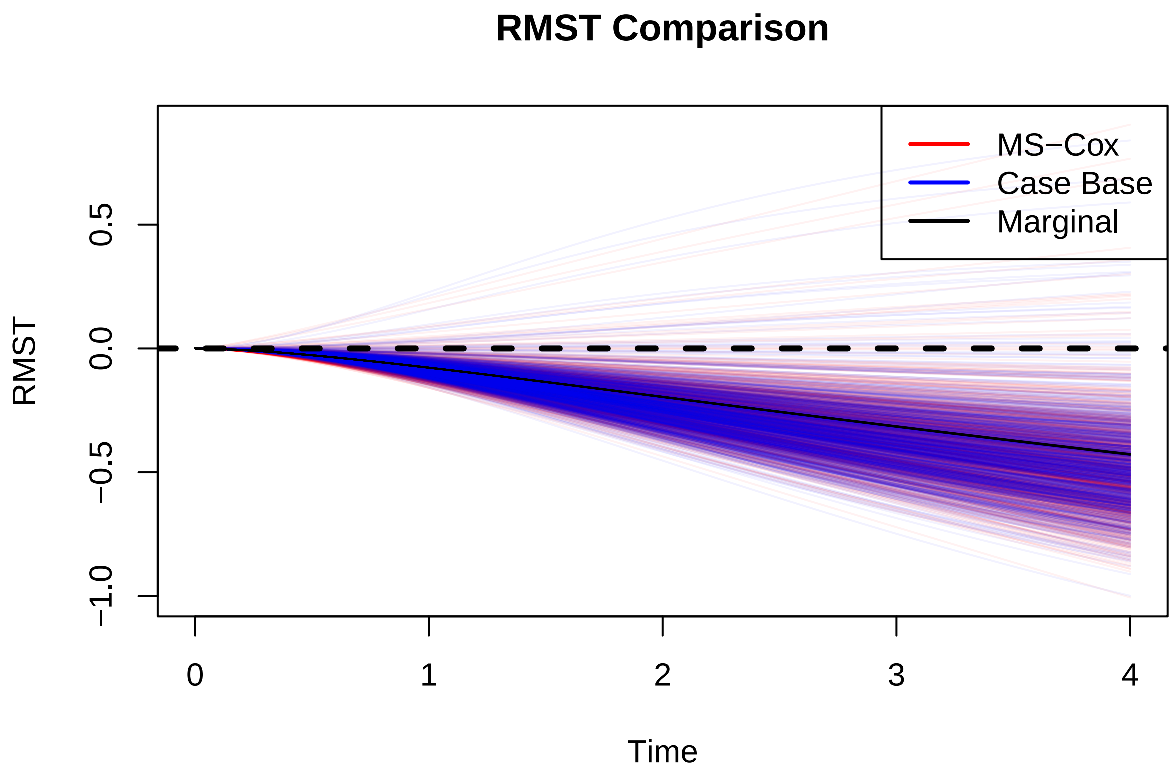
Run Simulation Study: Setting 6

```
run.complete.ss(
  n6, beta.A6,
  censoring.limit.6,
  true.estimates.moderate,
  generate.plots = TRUE
)
```




MS−Cox RMST
MS−Cox
Marginal
RMST
0.5
0.0
−0.5
−1.0
0
1
2
3
4
Time

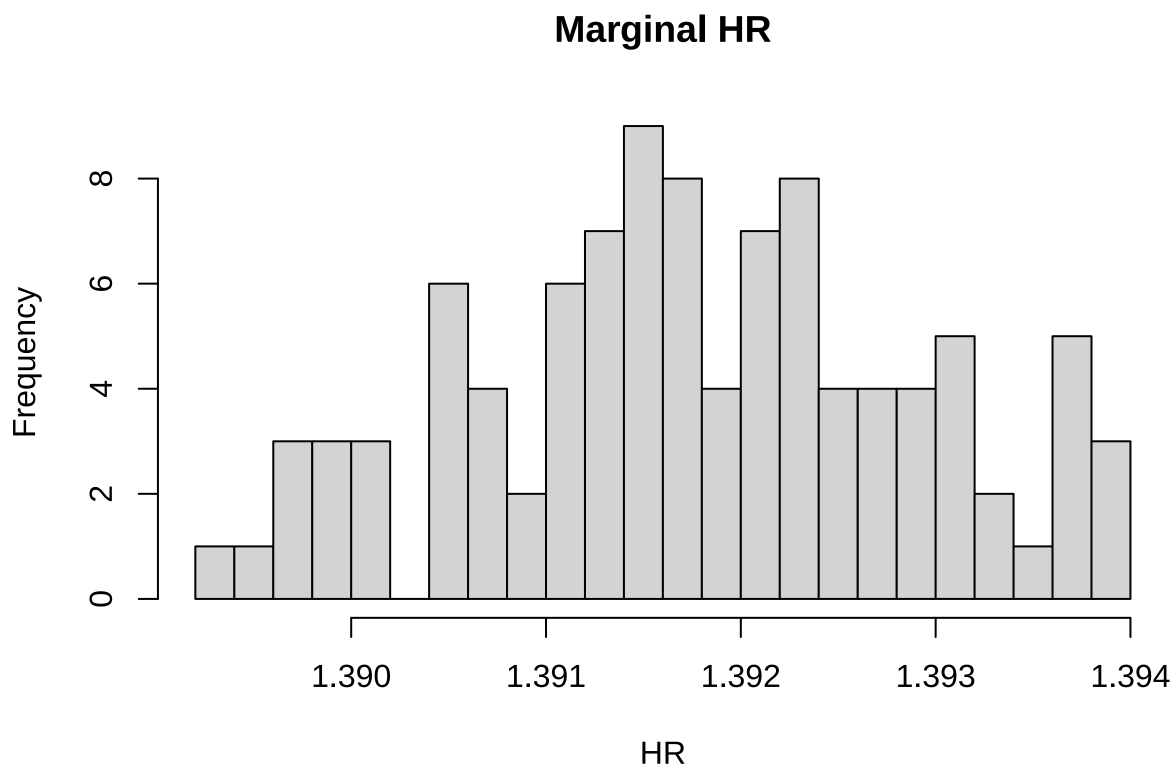

Case Base RMST
Case Base
Marginal
RMST
0.5
0.0
−0.5
−1.0
0
1
2
3
4
Time

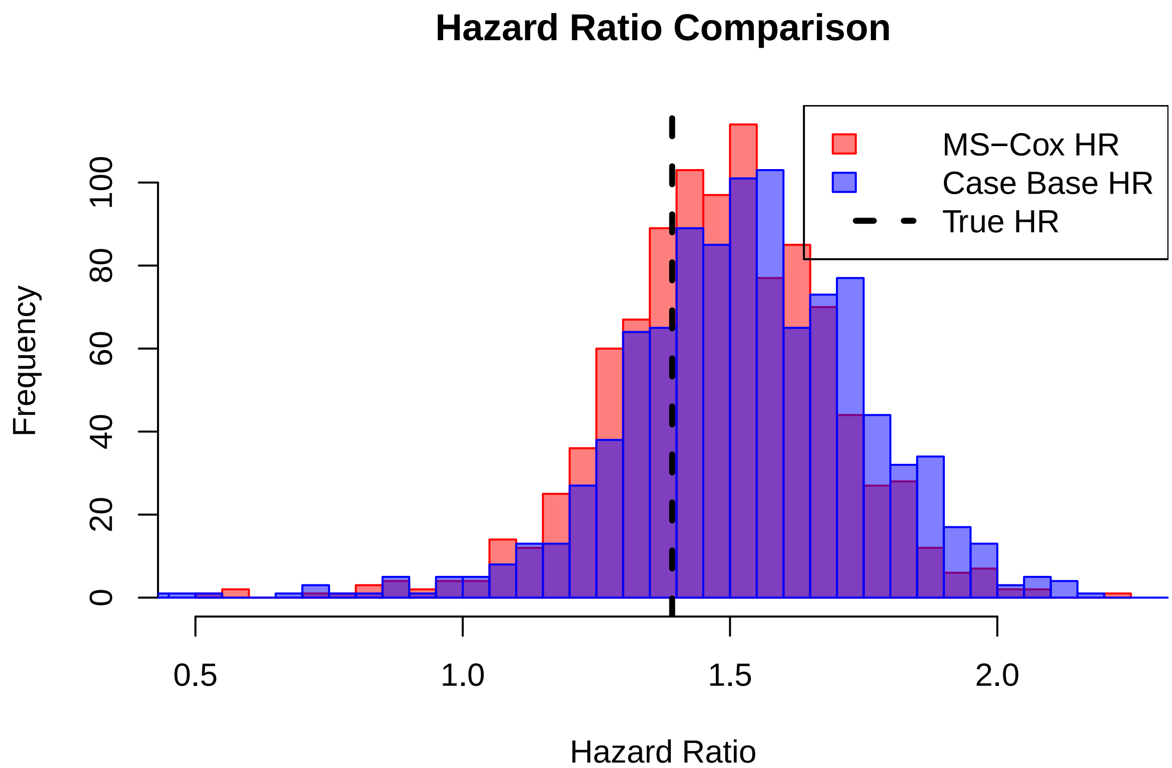



```
##
## ========================================
##
## True Mean Restricted Mean Survival Time
## ========================================
## 1 : -0.07760737
## 2 : -0.1956718
## 3 : -0.3150435
## 4 : -0.4276574
##
##
## ========================================
## MS-Cox RMST
## ========================================
##
## Restricted Mean Survival Time Summary
## ----------------------------------------
##
## Tau = 1
## ----------------------------------------
## Mean RMST: -0.08708602
## Bias: -0.009478645  | MC-SE: 0.001082281
## RMSE: 0.03549656
## Mean Estimated SE: 0.03860898  | Empirical SD: 0.03422473
## Coverage: 0.985  | MC-SE: 0.003843826
##
```

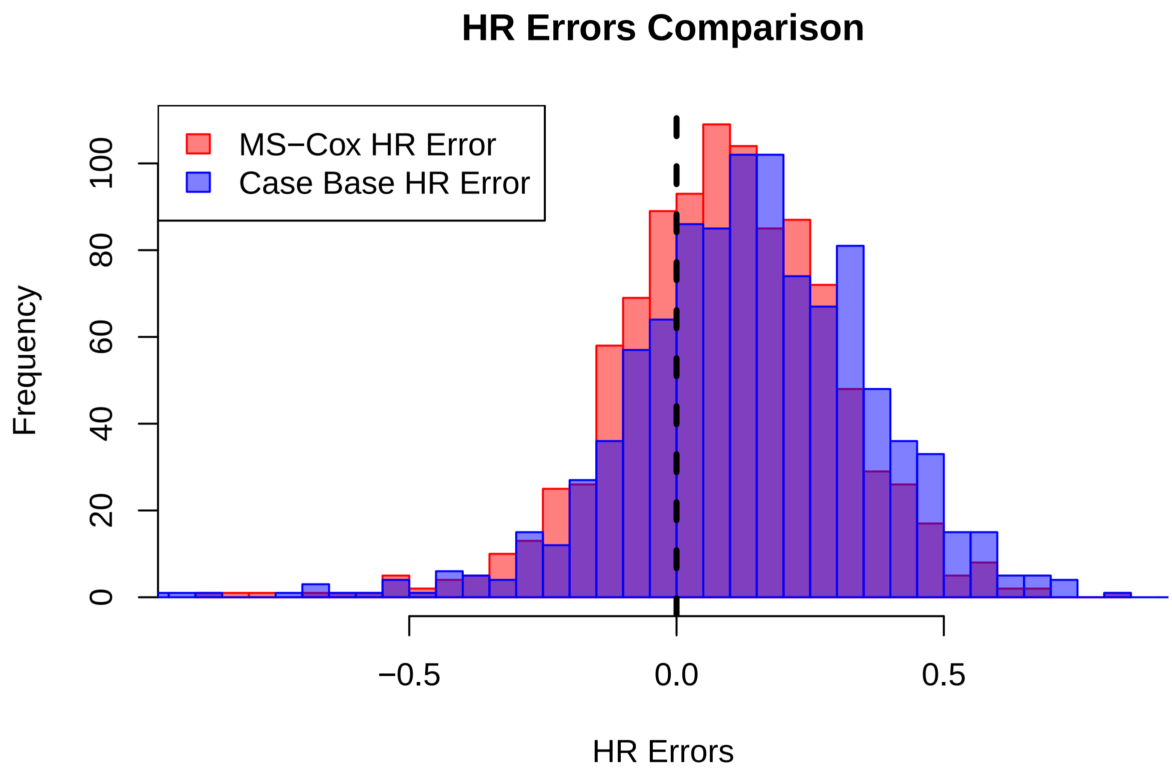
```
## Tau = 2
## ----------------------------------------
## Mean RMST: -0.2225206
## Bias: -0.02684885  | MC-SE: 0.002744671
## RMSE: 0.09081049
## Mean Estimated SE: 0.09321134  | Empirical SD: 0.08679411
## Coverage: 0.968  | MC-SE: 0.005565609
##
## Tau = 3
## ----------------------------------------
## Mean RMST: -0.3599376
## Bias: -0.04489409  | MC-SE: 0.004432866
## RMSE: 0.1471262
## Mean Estimated SE: 0.1450531  | Empirical SD: 0.1401795
## Coverage: 0.949  | MC-SE: 0.006956939
##
## Tau = 4
## ----------------------------------------
## Mean RMST: -0.4938451
## Bias: -0.06618775  | MC-SE: 0.006083219
## RMSE: 0.2033454
## Mean Estimated SE: 0.1941676  | Empirical SD: 0.1923683
## Coverage: 0.938  | MC-SE: 0.007626008
##
##
## ========================================
## Case Base RMST
## ========================================
##
## Restricted Mean Survival Time Summary
## ----------------------------------------
##
## Tau = 1
## ----------------------------------------
## Mean RMST: -0.08115419
## Bias: -0.003546819  | MC-SE: 0.001062881
## RMSE: 0.03378117
## Mean Estimated SE: 0.02895153  | Empirical SD: 0.03361126
## Coverage: 0.937  | MC-SE: 0.007683163
##
## Tau = 2
## ----------------------------------------
## Mean RMST: -0.2275889
## Bias: -0.03191718  | MC-SE: 0.002881042
## RMSE: 0.09649254
## Mean Estimated SE: 0.08220987  | Empirical SD: 0.09110655
## Coverage: 0.9  | MC-SE: 0.009486833
##
## Tau = 3
## ----------------------------------------
## Mean RMST: -0.3671774
## Bias: -0.05213392  | MC-SE: 0.004582498
## RMSE: 0.1539358
## Mean Estimated SE: 0.1352033  | Empirical SD: 0.1449113
```

```
## Coverage: 0.91  | MC-SE: 0.009049862
##
## Tau = 4
## ----------------------------------------
## Mean RMST: -0.4788787
## Bias: -0.0512213  | MC-SE: 0.005959359
## RMSE: 0.1951975
## Mean Estimated SE: 0.1801003  | Empirical SD: 0.1884515
## Coverage: 0.931  | MC-SE: 0.008014924
```





Hazard Ratio Comparison
MS−Cox HR
Case Base HR
True HR
Frequency
0
20
40
60
80
100
0.5
1.0
1.5
2.0
Hazard Ratio




```
##
## ========================================
## True HR: 1.391778
## ========================================
##
## ========================================
## MS-Cox
## ========================================
##
## Hazard Ratio Summary
## ----------------------------------------
## Mean HR: 1.466587
## Bias: 0.07480867  | MC-SE: 0.006540331
## MSE: 0.05093354
## RMSE: 0.2256846
## Mean Estimated SE: 0.1505033  | Empirical SD: 0.2068234
## Type I Error/Power: 0.761
## Coverage: 0.936  | MC-SE: 0.007739767
##
##
## ========================================
## Case Base
## ========================================
##
## Hazard Ratio Summary
## ----------------------------------------
```

```
## Mean HR: 1.513519
## Bias: 0.1217408  | MC-SE: 0.007277966
## MSE: 0.07285053
## RMSE: 0.2699084
## Mean Estimated SE: 0.1555925  | Empirical SD: 0.2301495
## Type I Error/Power: 0.791
## Coverage: 0.89  | MC-SE: 0.009894443
```